

If the star is collapsing,  $\epsilon_{\text{gravity}}$  will be positive; if it is expanding,  $\epsilon_{\text{gravity}}$  will be negative. Thus, as the star contracts, its entropy decreases. This is not a violation of the second law of thermodynamics, which states that the entropy of a *closed* system must always remain the same (reversible process) or increase (irreversible process). Since a star is not a closed system, its entropy may decrease locally while the entropy of the remainder of the universe increases by a greater amount. The entropy is carried out of the star by photons and neutrinos.

When changes in the structure of the star are sufficiently rapid that accelerations can no longer be neglected, Eq. (10.6) must be replaced by the exact expression, Eq. (10.5). Such a situation can occur during a supernova explosion or during stellar pulsations. The effect of the acceleration term in stellar pulsations will be discussed in Chapter 14.

### The Constitutive Relations

The basic stellar structure equations [(10.6), (10.7), (10.36), (10.68), and (10.89)] require information concerning the physical properties of the matter from which the star is made. The required conditions are the equations of state of the material and are collectively referred to as **constitutive relations**. Specifically, we need relationships for the pressure, the opacity, and the energy generation rate, in terms of fundamental characteristics of the material: the density, temperature, and composition. In general,

$$P = P(\rho, T, \text{composition}) \quad (10.103)$$

$$\bar{\kappa} = \bar{\kappa}(\rho, T, \text{composition}) \quad (10.104)$$

$$\epsilon = \epsilon(\rho, T, \text{composition}) \quad (10.105)$$

The pressure equation of state can be quite complex in the deep interiors of certain classes of stars, where the density and temperature can become extremely high. However, in most situations, the ideal gas law, combined with the expression for radiation pressure, is a good first approximation, particularly when the variation in the mean molecular weight with composition and ionization is properly calculated. The pressure equation of state developed earlier (Eq. 10.20) includes both the ideal gas law and radiation pressure.

The opacity of the stellar material cannot be expressed exactly by a single formula. Instead, it is calculated explicitly for various compositions at specific densities and temperatures and presented in tabular form. Stellar structure codes either interpolate in a density–temperature grid to obtain the opacity for the specified conditions, or, alternatively, use a “fitting function”, based on the tabulated values. A similar situation also occurs for accurate calculations of the pressure equation of state. Although no accurate fitting function can be constructed to account for bound–bound opacities, approximate expressions for bound–free, free–free, electron scattering, and  $\text{H}^-$  ion opacities were presented in Section 9.2; see Eqs. (9.22), (9.23), (9.27), and (9.28), respectively.

To calculate the nuclear energy generation rate, we can use formulas such as those presented in Section 10.3 for the pp chain (Eq. 10.46) and the CNO cycle (Eq. 10.58). In more sophisticated calculations, *reaction networks* are employed that yield individual reaction rates for each step of a process and equilibrium abundances for each isotope in the mixture.

### Boundary Conditions

The actual solution of the stellar structure equations, including the constitutive relations, requires appropriate **boundary conditions** that specify physical constraints to the mathematical equations. Boundary conditions play the essential role of defining the limits of integration. The central boundary conditions are fairly obvious—namely that the interior mass and luminosity must approach zero at the center of the star, or

$$\left. \begin{array}{l} M_r \rightarrow 0 \\ L_r \rightarrow 0 \end{array} \right\} \text{ as } r \rightarrow 0. \quad (10.106)$$

This simply means that the star is physically realistic and does not contain a hole, a core of negative luminosity, or central points of infinite  $\rho$  or  $\epsilon$ !

A second set of boundary conditions is required at the surface of the star. The simplest set of assumptions is that the temperature, pressure, and density all approach zero at some surface value for the star's radius,  $R_\star$ , or

$$\left. \begin{array}{l} T \rightarrow 0 \\ P \rightarrow 0 \\ \rho \rightarrow 0 \end{array} \right\} \text{ as } r \rightarrow R_\star. \quad (10.107)$$

Strictly, the conditions of Eqs. (10.107) will never be obtained in a real star (as is obviously the case for the temperature). Therefore, it is often necessary to use more sophisticated surface boundary conditions, such as when the star being modeled has an extended atmosphere or is losing mass, as most stars do.

### The Vogt–Russell Theorem

Given the basic stellar structure equations, constitutive relations, and boundary conditions, we can now specify the type of star to be modeled. As can be seen by examination of Eq. (10.6), the pressure gradient at a given radius is dependent on the interior mass and the density. Similarly, the radiative temperature gradient (Eq. 10.68) depends on the local temperature, density, opacity, and interior luminosity, while the luminosity gradient is a function of the density and energy generation rate. The pressure, opacity, and energy generation rate in turn depend explicitly on the density, temperature, and composition at that location. If the interior mass at the surface of the star (i.e., the entire stellar mass) is specified, along with the composition, surface radius, and luminosity, application of the surface boundary conditions allows for a determination of the pressure, interior mass, temperature, and interior luminosity at an infinitesimal distance  $dr$  below the surface of the star.<sup>23</sup> Continuing this *numerical integration* of the stellar structure equations to the center of the star must result agreement with the central boundary conditions (Eqs.

<sup>23</sup>It is also necessary to specify the average density over that distance. Since  $\rho$  is assumed to be zero at the surface, and since it depends explicitly on the pressure and temperature, which are also assumed to be zero at the surface and are initially unknown below the surface, an immediate difficulty arises; the right-hand sides of Eqs. (10.6) and (10.68) are *zero*, so  $P$  and  $T$  never increase from their surface values! One solution to this initial-step problem is outlined in Appendix L. More sophisticated solutions require an iterative procedure, continually correcting previous estimates until a self-consistent answer is obtained to within some specified level of accuracy.

10.106). Since the values of the various gradients are directly related to the composition of the star, it is not possible to specify any arbitrary combination of surface radius and luminosity after the mass and composition have been selected. This set of constraints is known as the **Vogt–Russell theorem**:

The mass and the composition structure throughout a star uniquely determine its radius, luminosity, and internal structure, as well as its subsequent evolution.

*The dependence of a star’s evolution on mass and composition is a consequence of the change in composition due to nuclear burning.*<sup>24</sup> The statement of the Vogt–Russell “theorem” given here is somewhat misleading since there are other parameters that can influence stellar interiors, such as magnetic fields and rotation. However, these parameters are assumed to have little effect in most stars and will not be discussed further.<sup>25</sup>

### Numerical Modeling of the Stellar Structure Equations

With the exception of a special family of approximate solutions to the stellar structure equations known as *polytropes* (to be discussed on page 334), the system of differential equations, along with their constitutive relations, cannot be solved analytically. Instead, as already mentioned, it is necessary to integrate the system of equations numerically. This is accomplished by approximating the differential equations by *difference equations*—by replacing  $dP/dr$  by  $\Delta P/\Delta r$ , for instance. The star is then imagined to be constructed of spherically symmetric **shells**, as in Fig. 10.11, and the “integration” is carried out from some initial radius in finite steps by specifying some increment  $\delta r$ .<sup>26</sup> It is then possible to increment each of the fundamental physical parameters through successive applications of the difference equations. For instance, if the pressure in zone  $i$  is given by  $P_i$ , then the pressure in the next deepest zone,  $P_{i+1}$ , is found from

$$P_{i+1} = P_i + \frac{\Delta P}{\Delta r} \delta r,$$

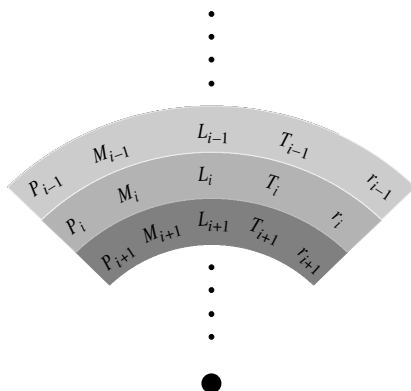
where  $\delta r$  is negative.

The numerical integration of the stellar structure equations may be carried out from the surface toward the center, from the center toward the surface, or, as is often done, in both directions simultaneously. If the integration is carried out in both directions, the solutions will meet at some *fitting point* where the variables must vary smoothly from one solution to the other. This last approach is frequently taken because the most important physical processes in the outer layers of stars generally differ from those in the deep interiors. The transfer of radiation through optically thin zones and the ionization of hydrogen and helium

<sup>24</sup>In this sense, Eq. (10.36) does contain an implicit time dependence due to stellar nucleosynthesis.

<sup>25</sup>Even without the complications of magnetic fields and rotation, the Vogt–Russell “theorem” can be violated in certain special circumstances. However, an actual star (as opposed to a theoretical model) would probably adopt one unique structure as a consequence of its evolutionary history. In this sense, the Vogt–Russell “theorem” should be considered a general rule rather than a rigorous law.

<sup>26</sup>Codes that treat the radius as an independent variable are called **Eulerian** codes. **Lagrangian** codes treat the mass as an independent variable. In the Lagrangian formulation, the differential equations are rewritten using Eq. (10.7); the hydrostatic equilibrium equation can be written in the form  $dP/dM$ , for instance.



**FIGURE 10.11** Zoning in a numerical stellar model. The star is assumed to be constructed of spherically symmetric mass shells, with the physical parameters associated with each zone being specified by the stellar structure equations, the constitutive relations, the boundary conditions, and the star’s mass and composition. In research-quality codes some quantities are specified in the middle of mass shells (e.g.,  $P$  and  $T$ ), whereas others are associated with the interfaces between shells (e.g.,  $r$ ,  $M_r$ , and  $L_r$ ).

occur close to the surface, while nuclear reactions occur near the center. By integrating in both directions, it is possible to decouple these processes somewhat, simplifying the problem.

Simultaneously matching the surface and central boundary conditions for a desired stellar model usually requires several iterations before a satisfactory solution is obtained. If the surface-to-center and center-to-surface integrations do not agree at the fitting point, the starting conditions must be changed. This is accomplished in a series of attempts, called *iterations*, where the initial conditions of the next integration are estimated from the outcome of the previous integration. A process of successive iterations is also necessary if the star is integrated from the surface to the center or from the center to the surface; in these cases the fitting points are simply the center and surface, respectively.

A very simple stellar structure code (called StatStar) is presented in Appendix L. StatStar integrates the stellar structure equations developed in this chapter in their time-independent form from the outside of the star to the center using the appropriate constitutive relations; it also assumes a constant (or homogeneous) composition throughout. Many of the sophisticated numerical techniques present in research codes have been neglected so that the basic elements of stellar model building can be more easily understood, as have the detailed calculations of the pressure equation of state and the opacity. The complex formalism of the mixing-length theory has also been left out in favor of the simplifying assumption of adiabatic convection. Despite these approximations, very reasonable models may be obtained for stars lying on the main sequence of the H–R diagram.

### Polytropic Models and the Lane–Emden Equation

As we mentioned previously, it is not generally possible to solve the system of stellar structure equations and their associated constitutive relations analytically; we must employ numerical solutions to “build” stellar models. However, under very special and restrictive

situations, it is possible to find analytic solutions to a subset of the equations. The first work in this area was carried out by J. Homer Lane (1819–1880), who wrote a paper on the equilibrium of stellar configurations in the *American Journal of Science* in 1869. That work was later extended significantly by Robert Emden (1862–1940). Today, the famous equation that helps us describe analytical stellar models is referred to as the *Lane–Emden equation*.

To understand the motivation of developing the Lane–Emden equation, note that careful inspection of the stellar structure equations shows that the mechanical equations of stellar structure (Eqs. 10.6 and 10.7) could be solved simultaneously without reference to the energy equations (10.36, and either 10.68 or 10.89) if only a simple relationship existed between pressure and density. Of course, as we have seen, such a simple relationship does not generally exist; normally, temperature and composition must also enter into the pressure equation of state, often in a complicated way. However, under certain circumstances, such as for an adiabatic gas (see Eq. 10.86), the pressure can be written explicitly in terms of the density alone. Hypothetical stellar models in which the pressure depends on density in the form  $P = K\rho^\gamma$  are known as **polytropes**. The development of polytropic models is well worth the effort since their relative simplicity allows us to gain some insight into stellar structure without all of the complications inherent in full-blown numerical models.

To derive the Lane–Emden equation, we begin with the equation for hydrostatic equilibrium, Eq. (10.6). Rewriting the equation and taking the radial derivative of both sides gives

$$\frac{d}{dr} \left( \frac{r^2}{\rho} \frac{dP}{dr} \right) = -G \frac{dM_r}{dr}.$$

We immediately see that Eq. (10.7) can be used to eliminate the mass gradient. Substituting, we get

$$\frac{d}{dr} \left( \frac{r^2}{\rho} \frac{dP}{dr} \right) = -G(4\pi r^2 \rho)$$

or

$$\frac{1}{r^2} \frac{d}{dr} \left( \frac{r^2}{\rho} \frac{dP}{dr} \right) = -4\pi G\rho. \quad (10.108)$$

As an aside, it is worth pointing out here that Eq. (10.108) is actually a slightly camouflaged form of a very well-studied differential equation known as *Poisson's equation*. It is left as an exercise to show that Eq. (10.108) can be rewritten in the form

$$\frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{d\Phi_g}{dr} \right) = 4\pi G\rho, \quad (10.109)$$

which is the spherically symmetric form of Poisson's equation for the gravitational potential energy per unit mass,  $\Phi_g \equiv U_g/m$ .<sup>27</sup>

<sup>27</sup>Poisson's equation shows up frequently in physics. For example, Gauss' Law, one of Maxwell's equations of electromagnetic theory, can be reformulated into Poisson's equation by replacing the electric field vector with the negative of the gradient of the electrostatic potential.

To solve Eq. (10.108), we now employ the relationship  $P(\rho) = K\rho^\gamma$ , where  $K$  and  $\gamma > 0$  are constants. This functional form of the pressure equation is known generally as a polytropic equation of state. Substituting, taking the appropriate derivative, and simplifying, we have

$$\frac{\gamma K}{r^2} \frac{d}{dr} \left[ r^2 \rho^{\gamma-2} \frac{d\rho}{dr} \right] = -4\pi G\rho.$$

It is customary to rewrite the expression slightly by letting  $\gamma \equiv (n+1)/n$ , where  $n$  is historically referred to as the *polytropic index*. Then

$$\left( \frac{n+1}{n} \right) \frac{K}{r^2} \frac{d}{dr} \left[ r^2 \rho^{(1-n)/n} \frac{d\rho}{dr} \right] = -4\pi G\rho.$$

In order to simplify the last expression somewhat, it is now useful to rewrite the equation in a dimensionless form. Expressing the density in terms of a scaling factor and a dimensionless function  $D(r)$ , let

$$\rho(r) \equiv \rho_c [D_n(r)]^n, \quad \text{where } 0 \leq D_n \leq 1.$$

(As you might suspect,  $\rho_c$  will turn out to be the central density of the polytropic stellar model.) Again substituting and simplifying, we arrive at

$$\left[ (n+1) \left( \frac{K\rho_c^{(1-n)/n}}{4\pi G} \right) \right] \frac{1}{r^2} \frac{d}{dr} \left[ r^2 \frac{dD_n}{dr} \right] = -D_n^n.$$

Careful study of our last equation reveals that the collective constant in square brackets has the units of distance squared. Defining

$$\lambda_n \equiv \left[ (n+1) \left( \frac{K\rho_c^{(1-n)/n}}{4\pi G} \right) \right]^{1/2}$$

and introducing the dimensionless independent variable  $\xi$  via

$$r \equiv \lambda_n \xi,$$

we finally arrive at

$$\boxed{\frac{1}{\xi^2} \frac{d}{d\xi} \left[ \xi^2 \frac{dD_n}{d\xi} \right] = -D_n^n,} \quad (10.110)$$

which is the famous **Lane–Emden equation**.

Solving Eq. (10.110) for the dimensionless function  $D_n(\xi)$  in terms of  $\xi$  for a specific polytropic index  $n$  leads directly to the profile of density with radius  $\rho_n(r)$ . The polytropic

equation of state  $P_n(r) = K\rho_n^{(n+1)/n}$  provides the pressure profile. In addition, if the ideal gas law and radiation pressure are assumed for constant composition (Eq. 10.20), then the temperature profile,  $T(r)$ , is also obtained.

In order to actually solve this second-order differential equation, it is necessary to impose two boundary conditions (which effectively specify the two constants of integration). Assuming that the “surface” of the star is that location where the pressure goes to zero (and correspondingly the density of the gas also goes to zero), then

$$D_n(\xi_1) = 0 \text{ specifies the surface at } \xi = \xi_1,$$

where  $\xi_1$  is the location of the first zero of the solution.

Next consider the center of the star. If  $r = \delta$  represents a distance infinitesimally close to the center of the star, then the mass contained within a volume of radius  $\delta$  is given by

$$M_r = \frac{4\pi}{3} \bar{\rho} \delta^3$$

where  $\bar{\rho}$  is the average density of the gas within the radius  $\delta$ . Substituting into the equation for hydrostatic equilibrium, Eq. (10.6), we have

$$\frac{dP}{dr} = -G \frac{M_r \rho}{r^2} = -\frac{4\pi}{3} G \bar{\rho}^2 \delta \rightarrow 0 \text{ as } \delta \rightarrow 0.$$

Since  $P = K\rho^{(n+1)/n}$ , this implies that

$$\frac{d\rho}{dr} \rightarrow 0 \text{ as } r \rightarrow 0,$$

which immediately leads to the central boundary condition

$$\frac{dD_n}{d\xi} = 0 \text{ at } \xi = 0.$$

In addition, in order for  $\rho_c$  to represent the central density of the star, it is also necessary that  $D_n(0) = 1$  (this condition isn't strictly a boundary condition, it simply *normalizes* the density scaling function,  $D_n$ ).

With the boundary conditions specified, it is now possible to compute the total mass of a star of a specific polytropic index. From Eq. (10.7),

$$M = 4\pi \int_0^R r^2 \rho \, dr,$$

where  $R = \lambda_n \xi_1$  represents the radius of the star. Rewriting in terms of the dimensionless quantities yields

$$M = 4\pi \int_0^{\xi_1} \lambda_n^2 \xi^2 \rho_c D_n^n d(\lambda_n \xi),$$

or

$$M = 4\pi\lambda_n^3\rho_c \int_0^{\xi_1} \xi^2 D_n^n d\xi.$$

Although this expression could be integrated directly with knowledge of  $D_n(\xi)$ , it can also be rewritten directly by noting, from the Lane–Emden equation and the central boundary condition, that

$$\xi^2 D_n^n = -\frac{d}{d\xi} \left[ \xi^2 \frac{dD_n}{d\xi} \right]$$

gives

$$M = -4\pi\lambda_n^3\rho_c \xi_1^2 \left. \frac{dD_n}{d\xi} \right|_{\xi_1},$$

where  $(dD_n/d\xi)|_{\xi_1}$  means that the derivative of  $D_n$  is evaluated at the surface.

Although the Lane–Emden equation is compact and elegant, it is important to bear in mind its many limitations. Recall that Eq. (10.110) contains no information about either energy transport or energy generation within a star; the equation only describes hydrostatic equilibrium and mass conservation, and then only within the highly idealized class of polytropic equations of state. Nevertheless, the Lane–Emden equation is capable of giving us some important insights into the structures of stars.

There are only three analytic solutions to the Lane–Emden equation, namely  $n = 0$ , 1, and 5. The  $n = 0$  solution is given by

$$D_0(\xi) = 1 - \frac{\xi^2}{6}, \quad \text{with } \xi_1 = \sqrt{6}.$$

It is left as an exercise for you to derive the  $n = 0$  solution. The solution for  $n = 1$  is the well-known “sinc” function

$$D_1(\xi) = \frac{\sin \xi}{\xi}, \quad \text{with } \xi_1 = \pi,$$

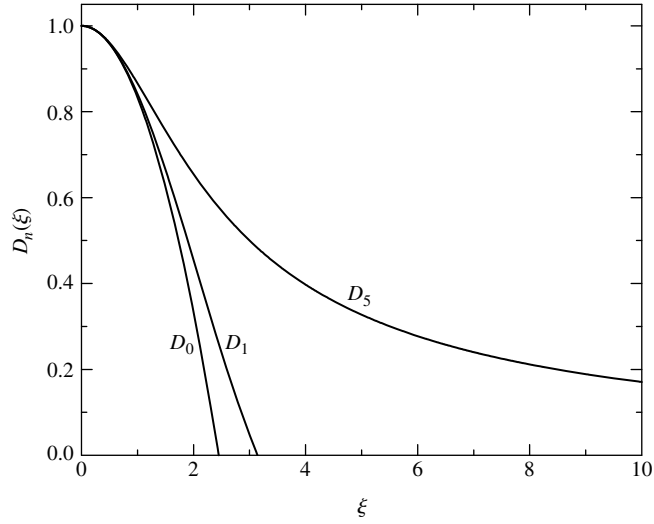
and the  $n = 5$  solution is given by

$$D_5(\xi) = [1 + \xi^2/3]^{-1/2}, \quad \text{with } \xi_1 \rightarrow \infty.$$

In the latter case you are asked to verify that although the radius of the star is infinite, the total mass of the star is actually finite. This is not the case for values of  $n > 5$ . Thus, the physical limits of  $n$  are constrained to the range  $0 \leq n \leq 5$ . Graphical representations of  $D_0$ ,  $D_1$ , and  $D_5$  are shown in Fig. 10.12.

This discussion of polytropes was originally motivated by the equation of state of an adiabatic gas. For the case of an ideal, monatomic gas,  $\gamma = 5/3$ , which implies that  $n = 1.5$ . In addition, as we shall see later, in Chapter 16 (see Eq. 16.12), certain extremely compressed stars in their final stage of evolution known as white dwarfs can also be described





**FIGURE 10.12** The analytic solutions to the Lane–Emden equation:  $D_0(\xi)$ ,  $D_1(\xi)$ , and  $D_5(\xi)$ .

by polytropes of index 1.5 (technically these are non-relativistic, completely degenerate stars). Although the important  $n = 1.5$  case cannot be solved analytically, it can be solved numerically.

Another important polytropic index is the  $n = 3$  “Eddington standard model” associated with a star in radiative equilibrium. To see how this model corresponds to radiative equilibrium, consider a polytrope that is supported by both an ideal gas and radiation pressure (see Eq. 10.20). If the total pressure at a certain location in the star is represented by  $P$ , and the contribution to that pressure due to an ideal gas is given by

$$P_g = \frac{\rho k T}{\mu m_H} = \beta P, \quad (10.111)$$

where  $0 \leq \beta \leq 1$ , then the contribution due to radiation pressure is

$$P_r = \frac{1}{3} a T^4 = (1 - \beta) P. \quad (10.112)$$

Since we are looking for a polytropic equation of state that can be expressed independent of temperature, we can combine the last two expressions to eliminate  $T$ . Solving for  $T$  in Eq. (10.111) and substituting into Eq. (10.112), we obtain

$$\frac{1}{3} a \left( \frac{\beta P \mu m_H}{\rho k} \right)^4 = (1 - \beta) P.$$

This leads immediately to an expression for the total pressure in terms of the density, namely

$$P = K \rho^{4/3} \quad (10.113)$$

where

$$K \equiv \left[ \frac{3(1-\beta)}{a} \right]^{1/3} \left( \frac{k}{\beta \mu m_H} \right)^{4/3}.$$

Since  $\gamma = 4/3$ , this implies that  $n = 3$ .<sup>28</sup>

Certainly the two most physically significant polytropic models correspond to  $n = 1.5$  and  $n = 3$ . Although neither model can be solved analytically, the use of computers and numerical integration algorithms allow us to explore their structure and behavior relatively easily. Careful study of these polytropes can yield important insights into the structures of more realistic, although significantly more complex stellar models.

## 10.6 ■ THE MAIN SEQUENCE

The analysis of stellar spectra tells us that the atmospheres of the vast majority of all stars are composed primarily of hydrogen, usually about 70% by mass ( $X \sim 0.7$ ), whereas the mass fraction of metals varies from near zero to approximately 3% ( $0 < Z < 0.03$ ). Assuming that the initial composition of a star is **homogeneous** (meaning that the composition is the same throughout), the first set of nuclear fusion reactions ought to be those that convert hydrogen into helium (the pp chains and/or the CNO cycle). Recall that these reactions occur at the lowest temperatures because the associated Coulomb barrier is lower than that for the burning of more massive nuclei. Consequently, the structure of a homogeneous, hydrogen-rich star ought to be strongly influenced by hydrogen nuclear burning deep within its interior.

Because of the predominance of hydrogen that initially exists in the core, and since hydrogen burning is a relatively slow process, the interior composition and structure of the star will change slowly. As we saw in Example 10.3.2, a rough estimate of the hydrogen-burning lifetime of the Sun is 10 billion years. Of course, the surface conditions will not be completely static. By the Vogt–Russell theorem, any change in composition or mass requires a readjustment of the effective temperature and luminosity; *the observational characteristic of the star must change as a consequence of the central nuclear reactions*. As long as changes in the core are slow, so are the evolutionary changes in the observed surface features.<sup>29</sup>

Since most stars have similar compositions, the structures of stars ought to vary smoothly with mass. Recall from Examples 10.1.1 and 10.2.1 that as the mass increases, the central pressure and the central temperature should increase. Therefore, for stars of low mass, the pp chain will dominate since less energy is required to initiate these reactions than the reactions of the CNO cycle. For high-mass stars, the CNO cycle will likely dominate because of its very strong temperature dependence.

<sup>28</sup>We will learn in Chapter 16 that stars supported solely by a fully relativistic, completely degenerate gas can also be described by a polytropic index of 3; see Eq. (16.15).

<sup>29</sup>Some short-period surface changes can occur that are essentially decoupled from the long-term variations in the core. Stellar pulsations require specific conditions to exist, but their timescales are usually much shorter than the nuclear timescale. These oscillations will be discussed in Chapter 14.

At some point, as progressively less massive stars are considered, the central temperature will diminish to the point where nuclear reactions are no longer able to stabilize a star against gravitational contraction. This has been shown to occur at approximately  $0.072 M_{\odot}$  for solar composition (the lower limit is slightly higher,  $0.09 M_{\odot}$ , for stars with virtually no metal content,  $Z \simeq 0$ ). At the other extreme, stars with masses greater than approximately  $90 M_{\odot}$  become subject to thermal oscillations in their centers that may produce significant variations in the nuclear energy generation rates over timescales as short as 8 hours.<sup>30</sup>

### The Eddington Luminosity Limit

Along with thermal oscillations, the stability of very massive stars is directly affected by their extremely high luminosities. As can be seen by Eq. (10.20), if the temperature is sufficiently high and the gas density is low enough, it is possible for radiation pressure to dominate over the gas pressure in certain regions of the star, a situation that can occur in the outer layers of very massive stars. In this case the pressure gradient is approximately given by Eq. (9.31). Combined with the relationship between radiant flux and luminosity (Eq. 3.2), the pressure gradient near the surface may be written as

$$\frac{dP}{dr} \simeq -\frac{\bar{\kappa}\rho}{c} \frac{L}{4\pi r^2}.$$

But hydrostatic equilibrium (Eq. 10.6) demands that the pressure gradient near the star's surface must also be given by

$$\frac{dP}{dr} = -G \frac{M\rho}{r^2},$$

where  $M$  is the star's mass. Combining, and solving for the luminosity, we have

$$L_{\text{Ed}} = \frac{4\pi Gc}{\bar{\kappa}} M. \quad (10.114)$$

$L_{\text{Ed}}$  is the maximum radiative luminosity that a star can have and still remain in hydrostatic equilibrium. If the luminosity exceeds  $L_{\text{Ed}}$ , mass loss must occur, driven by radiation pressure. This luminosity maximum, known as the **Eddington limit**, appears in a number of areas of astrophysics, including late stages of stellar evolution, novae, and the structure of accretion disks.

For our purposes, it is possible to make an estimate of the Eddington luminosity for stars on the upper end of the main sequence. The effective temperatures of these massive stars are in the range of 50,000 K, high enough that most of the hydrogen is ionized in their photospheres. Therefore, the major contribution to the opacity is from electron scattering, and we can replace  $\bar{\kappa}$  by Eq. (9.27). For  $X = 0.7$ , Eq. (10.114) becomes

$$L_{\text{Ed}} \simeq 1.5 \times 10^{31} \frac{M}{M_{\odot}} \text{ W} \quad \text{or} \quad \frac{L_{\text{Ed}}}{L_{\odot}} \simeq 3.8 \times 10^4 \frac{M}{M_{\odot}}.$$

<sup>30</sup>This so-called  $\epsilon$  *pulsation mechanism* will be discussed in Section 14.2.

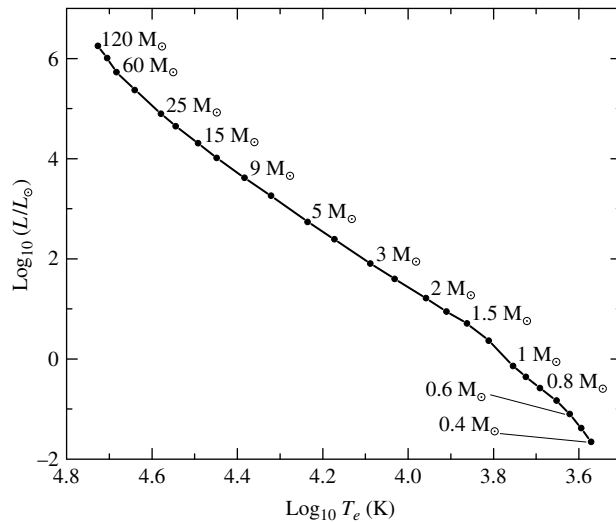
For a  $90 M_{\odot}$  star,  $L_{\text{Ed}} \simeq 3.5 \times 10^6 L_{\odot}$ , roughly three times the expected main-sequence value.

The fairly close correspondence between the theoretical and Eddington luminosities implies that the envelopes of massive main-sequence stars are loosely bound at best. In fact, observations of the few stars with masses estimated to be near  $100 M_{\odot}$  indicate that they are suffering from large amounts of mass loss and exhibit variability in their luminosities.

### Variations of Main-Sequence Stellar Parameters with Mass

From theoretical models that are computed in the mass range of hydrogen burning, it is possible to obtain a numerical relationship between  $M$  and  $L$  that agrees well with the observational mass–luminosity relation shown in Fig. 7.7. It is also possible to locate each of the models on a theoretical H–R diagram (see Fig. 10.13). By comparison with Fig. 8.13, it can be seen that stars undergoing hydrogen burning in their cores lie along the observational main sequence!

The range in main-sequence luminosities is from near  $5 \times 10^{-4} L_{\odot}$  to approximately  $1 \times 10^6 L_{\odot}$ , a variation of over nine orders of magnitude, while the masses change by only three orders of magnitude. Because of the enormous rate of energy output from upper main-sequence stars, they consume their core hydrogen in a much shorter period of time than do stars on the lower end of the main sequence. As a result, *main-sequence lifetimes decrease with increasing luminosity*. Estimates of the range of main-sequence lifetimes are left as an exercise.



**FIGURE 10.13** The locations of stellar models on a theoretical H–R diagram. The models were computed using the stellar structure equations and constitutive relations. (Data from Schaller, et al., *Astron. Astrophys. Suppl.*, 96, 269, 1992, and Charbonnel, et al., *Astron. Astrophys. Suppl.*, 135, 405, 1999.)

Effective temperatures are much less dependent on stellar mass. From approximately 1700 K for  $0.072 M_{\odot}$  stars to near 53,000 K for  $90 M_{\odot}$  stars, the increase in effective temperature amounts to a factor of only about 20. However, this variation is still large enough to dramatically change the stellar spectrum, since the dissociation energies of molecules and the ionization potentials of most elements lie within this range, as we demonstrated in Chapter 8. Consequently, by comparison with theoretical models, it is possible to correlate main-sequence masses with observed spectra.

The interior structure of stars along the main sequence also varies with mass, primarily in the location of convection zones. In the upper portion of the main sequence, where energy generation is due to the strongly temperature-dependent CNO cycle, convection is dominant in the core. This occurs because the rate of energy generation changes quickly with radius, and radiation is not efficient enough to transport all of the energy being released in nuclear reactions. Outside of the hydrogen-burning core, radiation is again capable of handling the flux, and convection ceases. As the stellar mass decreases, so does the central temperature and the energy output of the CNO cycle until, near  $1.2 M_{\odot}$ , the pp chain begins to dominate and the core becomes radiative. Meanwhile, near the surface of the star, as the effective temperature decreases with decreasing mass, the opacity increases, in part because of the location of the zone of hydrogen ionization (recall Fig. 9.10). The increase in opacity makes convection more efficient than radiation near the surfaces of stars with masses less than approximately  $1.3 M_{\odot}$ . This has the effect of creating convection zones near the surfaces of these stars. As we continue to move down the main sequence, the bottom of the surface convection zone lowers until the entire star becomes convective near  $0.3 M_{\odot}$ .

Through the use of the fundamental physical principles developed thus far in this text, we have been able to build realistic models of main-sequence stars and develop an understanding of their interiors. However, other stars remain on the observational H–R diagram that do not lie along the main sequence (see Fig. 8.13). By considering the changes in stellar structure that occur because of changes in composition due to nuclear burning (the Vogt–Russell theorem), it will become possible to explain their existence as well. The evolution of stars is discussed in Chapters 12 and 13.

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## PROBLEMS

- 10.1** Show that the equation for hydrostatic equilibrium, Eq. (10.6), can also be written in terms of the optical depth  $\tau$ , as

$$\frac{dP}{d\tau} = \frac{g}{\kappa}.$$

This form of the equation is often useful in building model stellar atmospheres.

- 10.2** Prove that the gravitational force on a point mass located anywhere inside a hollow, spherically symmetric shell is zero. Assume that the mass of the shell is  $M$  and has a constant density  $\rho$ . Assume also that the radius of the inside surface of the shell is  $r_1$  and that the radius of the outside surface is  $r_2$ . The mass of the point is  $m$ .
- 10.3** Assuming that 10 eV could be released by every atom in the Sun through chemical reactions, estimate how long the Sun could shine at its current rate through chemical processes alone. For simplicity, assume that the Sun is composed entirely of hydrogen. Is it possible that the Sun's energy is entirely chemical? Why or why not?
- 10.4** (a) Taking into consideration the Maxwell–Boltzmann velocity distribution, what temperature would be required for two protons to collide if quantum mechanical tunneling is neglected? Assume that nuclei having velocities ten times the root-mean-square (rms) value for the Maxwell–Boltzmann distribution can overcome the Coulomb barrier. Compare your answer with the estimated central temperature of the Sun.
- (b) Using Eq. (8.1), calculate the ratio of the number of protons having velocities ten times the rms value to those moving at the rms velocity.
- (c) Assuming (incorrectly) that the Sun is pure hydrogen, estimate the number of hydrogen nuclei in the Sun. Could there be enough protons moving with a speed ten times the rms value to account for the Sun's luminosity?
- 10.5** Derive the ideal gas law, Eq. (10.10). Begin with the pressure integral (Eq. 10.9) and the Maxwell–Boltzmann velocity distribution function (Eq. 8.1).
- 10.6** Derive Eq. (10.28) from Eq. (8.1).
- 10.7** By invoking the virial theorem (Eq. 2.46), make a crude estimate of an “average” temperature for the Sun. Is your result consistent with other estimates obtained in Chapter 10? Why or why not?
- 10.8** Show that the form of the Coulomb potential barrier penetration probability given by Eq. (10.31) follows directly from Eq. (10.30).
- 10.9** Prove that the energy corresponding to the Gamow peak is given by Eq. (10.34).
- 10.10** Calculate the ratio of the energy generation rate for the pp chain to the energy generation rate for the CNO cycle given conditions characteristic of the center of the present-day (evolved) Sun, namely  $T = 1.5696 \times 10^7$  K,  $\rho = 1.527 \times 10^5$  kg m<sup>-3</sup>,  $X = 0.3397$ , and  $X_{\text{CNO}} = 0.0141$ .<sup>31</sup> Assume that the pp chain screening factor is unity ( $f_{pp} = 1$ ) and that the pp chain branching factor is unity ( $\psi_{pp} = 1$ ).

<sup>31</sup>The interior values assumed here are taken from the solar model of Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.

**10.11** Beginning with Eq. (10.62) and writing the energy generation rate in the form

$$\epsilon(T) = \epsilon'' T_8^\alpha,$$

show that the temperature dependence for the triple alpha process, given by Eq. (10.63), is correct.  $\epsilon''$  is a function that is independent of temperature.

*Hint:* First take the natural logarithm of both sides of Eq. (10.62) and then differentiate with respect to  $\ln T_8$ . Follow the same procedure with your power law form of the equation and compare the results. You may want to make use of the relation

$$\frac{d \ln \epsilon}{d \ln T_8} = \frac{d \ln \epsilon}{\frac{1}{T_8} d T_8} = T_8 \frac{d \ln \epsilon}{d T_8}.$$

**10.12** The  $Q$  value of a reaction is the amount of energy released (or absorbed) during the reaction. Calculate the  $Q$  value for each step of the PP I reaction chain (Eqs. 10.37–10.39). Express your answers in MeV. The masses of  ${}^2_1\text{H}$  and  ${}^3_2\text{He}$  are 2.0141 u and 3.0160 u, respectively.

**10.13** Calculate the amount of energy released or absorbed in the following reactions (express your answers in MeV):

- (a)  ${}^{12}_6\text{C} + {}^{12}_6\text{C} \rightarrow {}^{24}_{12}\text{Mg} + \gamma$
- (b)  ${}^{12}_6\text{C} + {}^{12}_6\text{C} \rightarrow {}^{16}_8\text{O} + 2 {}^4_2\text{He}$
- (c)  ${}^{19}_9\text{F} + {}^1_1\text{H} \rightarrow {}^{16}_8\text{O} + {}^4_2\text{He}$

The mass of  ${}^{12}_6\text{C}$  is 12.0000 u, by definition, and the masses of  ${}^{16}_8\text{O}$ ,  ${}^{19}_9\text{F}$ , and  ${}^{24}_{12}\text{Mg}$  are 15.99491 u, 18.99840 u, and 23.98504 u, respectively. Are these reactions exothermic or endothermic?

**10.14** Complete the following reaction sequences. Be sure to include any necessary leptons.

- (a)  ${}^{27}_{14}\text{Si} \rightarrow {}^{27}_{13}\text{Al} + e^+ + ?$
- (b)  ${}^{27}_{13}\text{Al} + {}^1_1\text{H} \rightarrow {}^{24}_{12}\text{Mg} + {}^4_2\text{He}$
- (c)  ${}^{35}_{17}\text{Cl} + {}^1_1\text{H} \rightarrow {}^{36}_{18}\text{Ar} + ?$

**10.15** Prove that Eq. (10.83) follows from Eq. (10.82).

**10.16** Show that Eq. (10.109) can be obtained from Eq. (10.108).

**10.17** Starting with the Lane–Emden equation and imposing the necessary boundary conditions, prove that the  $n = 0$  polytrope has a solution given by

$$D_0(\xi) = 1 - \frac{\xi^2}{6}, \quad \text{with } \xi_1 = \sqrt{6}.$$

**10.18** Describe the density structure associated with an  $n = 0$  polytrope.

**10.19** Derive an expression for the total mass of an  $n = 5$  polytrope, and show that although  $\xi_1 \rightarrow \infty$ , the mass is finite.

- 10.20** (a) On the same graph, plot the density structure of stars of polytropic indices  $n = 0$ ,  $n = 1$ , and  $n = 5$ . *Hint:* You will want to plot  $\rho_n/\rho_c$  vs.  $r/\lambda_n$ .
- (b) What can you conclude about the concentration of density with radius for increasing polytropic index?
- (c) From the trend that you observe for the analytic solutions to the Lane–Emden equation, what would you expect regarding the density concentration of an adiabatically convective stellar model compared to a model in radiative equilibrium?



- (d) Explain your conclusion in part (c) in terms of the physical processes of convection and radiation.
- 10.21** Estimate the hydrogen-burning lifetimes of stars near the lower and upper ends of the main sequence. The lower end of the main sequence<sup>32</sup> occurs near  $0.072 M_{\odot}$ , with  $\log_{10} T_e = 3.23$  and  $\log_{10}(L/L_{\odot}) = -4.3$ . On the other hand, an  $85 M_{\odot}$  star<sup>33</sup> near the upper end of the main sequence has an effective temperature and luminosity of  $\log_{10} T_e = 4.705$  and  $\log_{10}(L/L_{\odot}) = 6.006$ , respectively. Assume that the  $0.072 M_{\odot}$  star is entirely convective so that, through convective mixing, all of its hydrogen, rather than just the inner 10%, becomes available for burning.
- 10.22** Using the information given in Problem 10.21, calculate the radii of a  $0.072 M_{\odot}$  star and a  $85 M_{\odot}$  star. What is the ratio of their radii?
- 10.23** (a) Estimate the Eddington luminosity of a  $0.072 M_{\odot}$  star and compare your answer to the main-sequence luminosity given in Problem 10.21. Assume  $\bar{\kappa} = 0.001 \text{ m}^2 \text{ kg}^{-1}$ . Is radiation pressure likely to be significant in the stability of a low-mass main-sequence star?
- (b) If a  $120 M_{\odot}$  star forms with  $\log_{10} T_e = 4.727$  and  $\log_{10}(L/L_{\odot}) = 6.252$ , estimate its Eddington luminosity, assuming the opacity is due to electron scattering. Compare your answer with the actual luminosity of the star.

## COMPUTER PROBLEMS

- 10.24** (a) Use a numerical integration algorithm such as the Euler method to compute the density profile for the  $n = 1.5$  and  $n = 3$  polytropes. Be sure to correctly incorporate the boundary conditions in your integrations.
- (b) Plot your results and compare them with the  $n = 0$ ,  $n = 1$ , and  $n = 5$  analytic models determined in Problem 10.20.
- 10.25** Verify that the basic equations of stellar structure [Eqs. (10.6), (10.7), (10.36), (10.68)] are satisfied by the  $1 M_{\odot}$  StatStar model available for download from the companion website; see Appendix L. This may be done by selecting two adjacent zones and numerically computing the derivatives on the left-hand sides of the equations, for example

$$\frac{dP}{dr} \simeq \frac{P_{i+1} - P_i}{r_{i+1} - r_i},$$

and comparing your results with results obtained from the right-hand sides using average values of quantities for the two zones [e.g.,  $M_r = (M_i + M_{i+1})/2$ ].

Perform your calculations for two adjacent shells at temperatures near  $5 \times 10^6 \text{ K}$ , and then compare your results for the left- and right-hand sides of each equation by determining relative errors. Note that the model assumes complete ionization everywhere and has the uniform composition  $X = 0.7$ ,  $Y = 0.292$ ,  $Z = 0.008$ . Your results on the left- and right-hand sides of the stellar structure equations will not agree exactly because StatStar uses a Runge–Kutta numerical algorithm that carries out intermediate steps not shown in the output file.

- 10.26** The companion website contains an example of a theoretical  $1.0 M_{\odot}$  main-sequence star produced by the stellar structure code StatStar, described in Appendix L. Using StatStar, build

<sup>32</sup>Data from Chabrier, et al., *Ap. J.*, 542, 464, 2000.

<sup>33</sup>Data from Schaller, et al., *Astron. Astrophys. Suppl. Ser.*, 96, 269, 1992.

a second main-sequence star with a mass of  $0.75 M_{\odot}$  that has a homogeneous composition of  $X = 0.7$ ,  $Y = 0.292$ , and  $Z = 0.008$ . For these values, the model's luminosity and effective temperature are  $0.189 L_{\odot}$  and  $3788.5 \text{ K}$ , respectively. Compare the central temperatures, pressures, densities, and energy generation rates between the  $1.0 M_{\odot}$  and  $0.75 M_{\odot}$  models. Explain the differences in the central conditions of the two models.

**10.27** Use the stellar structure code `StatStar` described in Appendix L, together with the theoretical `StatStar` H–R diagram and mass–effective temperature data provided on the companion website, to calculate a homogeneous, main-sequence model having the composition  $X = 0.7$ ,  $Y = 0.292$ , and  $Z = 0.008$ . (*Note:* It may be more illustrative to assign each student in the class a different mass for this problem so that the results can be compared.)

- (a) After obtaining a satisfactory model, plot  $P$  versus  $r$ ,  $M_r$  versus  $r$ ,  $L_r$  versus  $r$ , and  $T$  versus  $r$ .
- (b) At what temperature has  $L_r$  reached approximately 99% of its surface value? 50% of its surface value? Is the temperature associated with 50% of the total luminosity consistent with the rough estimate found in Eq. (10.27)? Why or why not?
- (c) What are the values of  $M_r/M_*$  for the two temperatures found in part (b)?  $M_*$  is the total mass of the stellar model.
- (d) If each student in the class calculated a different mass, compare the changes in the following quantities with mass:
  - (i) The central temperature.
  - (ii) The central density.
  - (iii) The central energy generation rate.
  - (iv) The extent of the central convection zone with mass fraction and radius.
  - (v) The effective temperature.
  - (vi) The radius of the star.
- (e) If each student in the class calculated a different mass:
  - (i) Plot each model on a graph of luminosity versus mass (i.e., plot  $L_*/L_{\odot}$  versus  $M_*/M_{\odot}$ ).
  - (ii) Plot  $\log_{10}(L_*/L_{\odot})$  versus  $\log_{10}(M_*/M_{\odot})$  for each stellar model.
  - (iii) Using an approximate power law relation of the form

$$L_*/L_{\odot} = (M_*/M_{\odot})^{\alpha},$$

find an appropriate value for  $\alpha$ .  $\alpha$  may differ for different compositions or vary somewhat with mass. This is known as the mass–luminosity relation (see Fig. 7.7).

**10.28** Repeat Problem 10.27 using the same mass but a different composition; assume  $X = 0.7$ ,  $Y = 0.290$ ,  $Z = 0.010$ .

- (a) For a given mass, which model ( $Z = 0.008$  or  $Z = 0.010$ ) has the higher central temperature? the greater central density?
- (b) Referring to the appropriate stellar structure equations and constitutive relations, explain your results in part (a).
- (c) Which model has the largest energy generation rate at the center? Why?
- (d) How do you account for the differences in effective temperature and luminosity between your two models?

## CHAPTER

# 11

## The Sun

- 11.1 *The Solar Interior*
- 11.2 *The Solar Atmosphere*
- 11.3 *The Solar Cycle*

### 11.1 ■ THE SOLAR INTERIOR

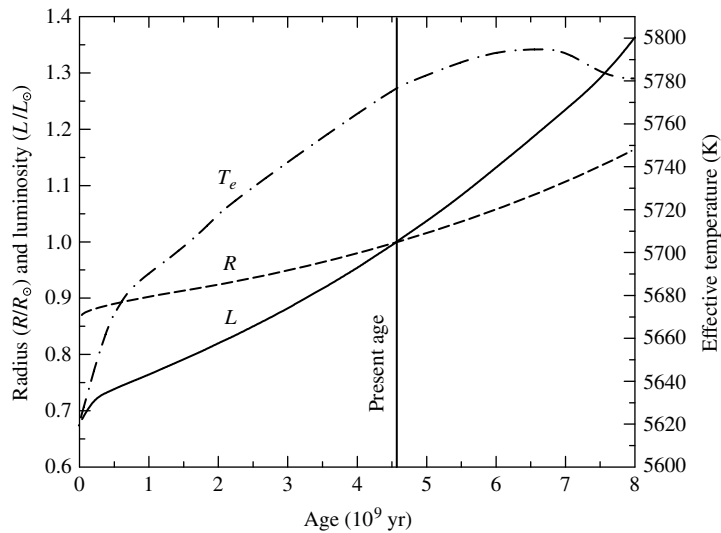
Over the last few chapters we have investigated the theoretical foundations of stellar structure, treating the star as being composed of an atmosphere and an interior. The distinction between the two regions is fairly nebulous. Loosely, the atmosphere is considered to be that region where the optical depth is less than unity and the simple approximation of photons *diffusing* through optically thick material is not justified (see Eq. 9.31). Instead, atomic line absorption and emission must be considered in detail in the stellar atmosphere. On the other hand, nuclear reaction processes deep in the stellar interior plays a crucial role in the star's energy output and its inevitable evolution.

Due to its proximity to us, the star for which we have the greatest amount of observational data is our Sun. From ground-based and space-based observatories, we are able to measure with high precision the composition of our Sun's surface; its luminosity, effective temperature, radius, magnetic fields, and rotation rates; the oscillation frequencies (vibrations) throughout its interior;<sup>1</sup> and the rate at which neutrinos are produced via nuclear reactions in its core. This tremendous wealth of information provides us with rigorous tests of our understanding of the physical processes operating within stellar atmospheres and interiors.

#### The Evolutionary History of the Sun

Based on its observed luminosity and effective temperature, our Sun is classified as a typical main-sequence star of spectral class G2 with a surface composition of  $X = 0.74$ ,  $Y = 0.24$ , and  $Z = 0.02$  (the mass fractions of hydrogen, helium, and metals, respectively). To understand how it has evolved to this point, recall that according to the Vogt–Russell theorem the mass and composition of a star dictate its internal structure. Our Sun has been converting hydrogen to helium via the pp chain during most of its lifetime, thereby changing its composition and its structure. By comparing the results of radioactive dating tests of Moon rocks and meteorites with stellar evolution calculations and the present-day

<sup>1</sup>Helioseismology, the study of solar oscillations, is discussed in Chapter 14.



**FIGURE 11.1** The evolution of the Sun on the main sequence. As a result of changes in its internal composition, the Sun has become larger and brighter. The solid line indicates its luminosity, the dashed line its radius, and the dash-dot line its effective temperature. The luminosity and radius curves are relative to present-day values. (Data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.)

observable Sun, the current age of the Sun is determined to be approximately  $4.57 \times 10^9$  yr.<sup>2</sup> Furthermore, as depicted in Fig. 11.1, since becoming a main-sequence star, the Sun's luminosity has increased nearly 48% (from  $0.677 L_{\odot}$ ) while its radius has increased 15% from an initial value of  $0.869 R_{\odot}$ .<sup>3</sup> The Sun's effective temperature has also increased from 5620 K to its present-day value of 5777 K (recall Example 3.4.2).

You may be wondering what impact this evolution has had on Earth. Interestingly, from a theoretical standpoint it is not at all clear how this change in solar energy output altered our planet during its history, primarily because of uncertainties in the behavior of the terrestrial environment. Understanding the complex interaction between the Sun and Earth involves the detailed calculation of convection in Earth's atmosphere, as well as the effects of the atmosphere's time-varying composition and the nature of the continually changing reflectivity, or **albedo**,<sup>4</sup> of Earth's surface.

### The Present-Day Interior Structure of the Sun

Consistent with the current age of the Sun, a **solar model** may be constructed for the present-day Sun using the physical principles discussed in preceding chapters. Table 11.1 gives the values of the central temperature, pressure, density, and composition for one such

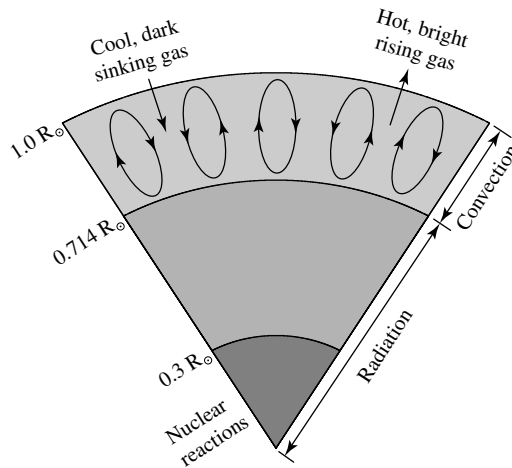
<sup>2</sup>Radioactive dating of the oldest known objects in the Solar System, calcium-aluminum-rich inclusions (CAIs) in meteorites, leads to a determination of the age of the Solar System of  $4.5672 \pm 0.0006$  Gyr.

<sup>3</sup>The data quoted here and in the following discussion are from the solar model of Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.

<sup>4</sup>Earth's albedo, the ratio of reflected to incident sunlight, is affected by the amount of surface water and ice.

**TABLE 11.1** Central Conditions in the Sun. (Data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.)

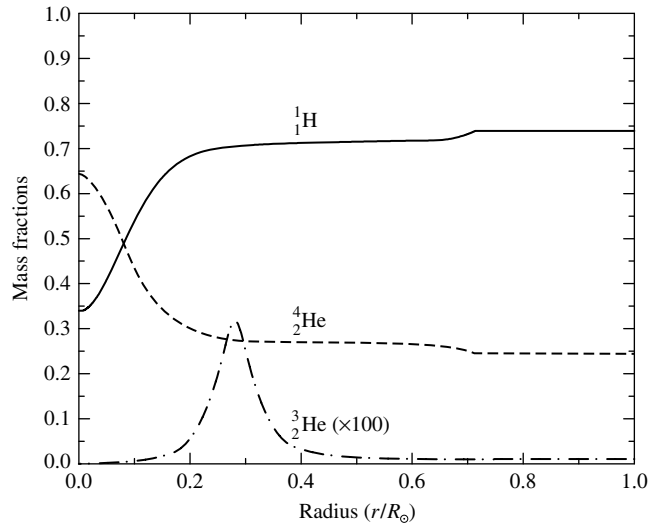
|             |                                         |
|-------------|-----------------------------------------|
| Temperature | $1.570 \times 10^7 \text{ K}$           |
| Pressure    | $2.342 \times 10^{16} \text{ N m}^{-2}$ |
| Density     | $1.527 \times 10^5 \text{ kg m}^{-3}$   |
| $X$         | 0.3397                                  |
| $Y$         | 0.6405                                  |

**FIGURE 11.2** A schematic diagram of the Sun's interior.

solar model, and a schematic diagram of the model is shown in Fig. 11.2. According to the evolutionary sequence leading to this model, during its lifetime the mass fraction of hydrogen ( $X$ ) in the Sun's center has decreased from its initial value of 0.71 to 0.34, while the central mass fraction of helium ( $Y$ ) has increased from 0.27 to 0.64. In addition, due to diffusive settling of elements heavier than hydrogen, the mass fraction of hydrogen near the surface has increased by approximately 0.03, while the mass fraction of helium has decreased by 0.03.

Because of the Sun's past evolution, its composition is no longer homogeneous but instead shows the influence of ongoing nucleosynthesis, surface convection, and elemental diffusion (settling of heavier elements). The composition structure of the Sun is shown in Fig. 11.3 for  ${}^1_1\text{H}$ ,  ${}^3_2\text{He}$ , and  ${}^4_2\text{He}$ . Since the Sun's primary energy production mechanism is the pp chain,  ${}^3_2\text{He}$  is an intermediate species in the reaction sequence. During the conversion of hydrogen to helium,  ${}^3_2\text{He}$  is produced and then destroyed again (see Fig. 10.8). At the top of the hydrogen-burning region where the temperature is lower,  ${}^3_2\text{He}$  is relatively more abundant because it is produced more easily than it is destroyed.<sup>5</sup> At greater depths, the higher temperatures allow the  ${}^3_2\text{He}$ – ${}^3_2\text{He}$  interaction to proceed more rapidly, and the  ${}^3_2\text{He}$

<sup>5</sup>Recall that much higher temperatures are required for helium–helium interactions than proton–proton interactions.



**FIGURE 11.3** The abundances of  ${}^1_1\text{H}$ ,  ${}^3_2\text{He}$ , and  ${}^4_2\text{He}$  as a function of radius for the Sun. Note that the abundance of  ${}^3_2\text{He}$  is multiplied by a factor of 100. (Data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.)

abundance again decreases (the temperature profile of the Sun is shown in Fig. 11.4). The slight ramp in the  ${}^1_1\text{H}$  and  ${}^4_2\text{He}$  curves near  $0.7 R_\odot$  reflects evolutionary changes in the position of the base of the surface convection zone, combined with the effects of elemental diffusion. Within the convection zone, turbulence results in essentially complete mixing and a homogeneous composition. The base of the present-day convection zone is at  $0.714 R_\odot$ .

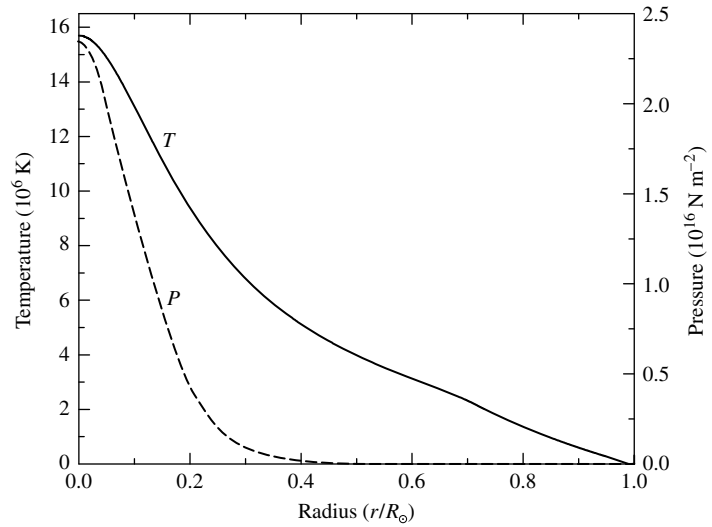
The largest contribution to the energy production in the Sun occurs at approximately one-tenth of the solar radius, as can be seen in the Sun's interior luminosity profile and the curve of its derivative with respect to radius (Fig. 11.5). If this result seems unexpected, consider that the mass conservation equation (Eq. 10.7),

$$\frac{dM_r}{dr} = 4\pi r^2 \rho,$$

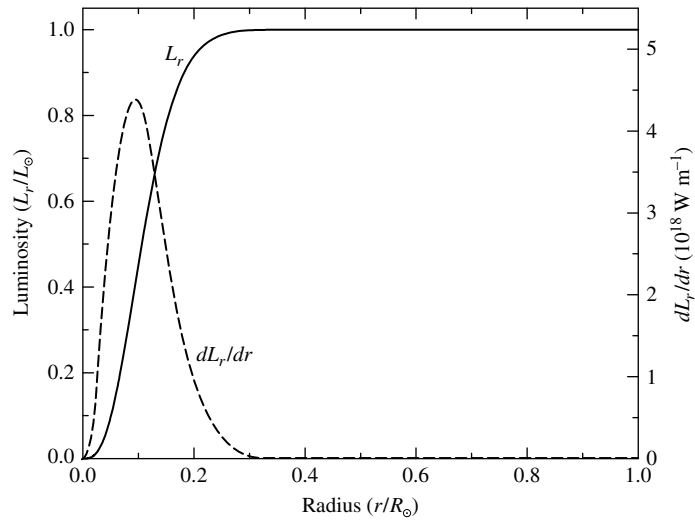
gives

$$dM_r = 4\pi r^2 \rho dr = \rho dV, \quad (11.1)$$

indicating that the amount of mass within a certain radius interval increases with radius simply because the volume of a spherical shell,  $dV = 4\pi r^2 dr$ , increases with  $r$  for a fixed choice of  $dr$ . Of course, the mass contained in the shell also depends on the density of the gas. Consequently, even if the amount of energy liberated per kilogram of material ( $\epsilon$ ) decreases steadily from the center outward, the largest contribution to the total luminosity will occur, not at the center, but in a shell that contains a significant amount of mass. In the case of the middle-aged Sun, the decrease in the amount of available hydrogen fuel at its center will also influence the location of the peak in the energy production region (see Eq. 10.46).

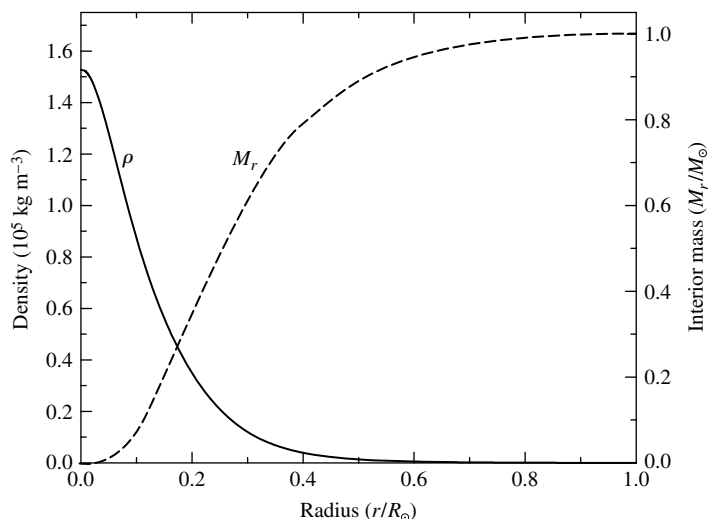


**FIGURE 11.4** The temperature and pressure profiles in the solar interior. (Data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.)



**FIGURE 11.5** The interior luminosity profile of the Sun and the derivative of the interior luminosity as a function of radius. (Data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.)

Figures 11.4 and 11.6 show just how rapidly the pressure and density change with radius in the Sun. These variations are forced on the solar structure by the condition of hydrostatic equilibrium (Eq. 10.6), the ideal gas law (Eq. 10.11), and the composition structure of the star. Of course, boundary conditions applied to the stellar structure equations require that both  $\rho$  and  $P$  become negligible at the surface (Eq. 10.107).



**FIGURE 11.6** The density profile and the interior mass of the Sun as a function of radius. (Data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001.)

Figure 11.6 also shows the interior mass ( $M_r$ ) as a function of radius. Notice that 90% of the mass of the star is located within roughly one-half of its radius. This should not come as a complete surprise since the density increases significantly as the center of the Sun is approached. Integration of the density function over the volume of the star from the center outward (i.e., the integration of Eq. 11.1) yields the interior mass function.

The question remains as to how the energy generated in the interior is transported outward. Recall that in Chapter 10 we determined a criterion for the onset of convection in stellar interiors, namely that the temperature gradient become superadiabatic (Eq. 10.94),

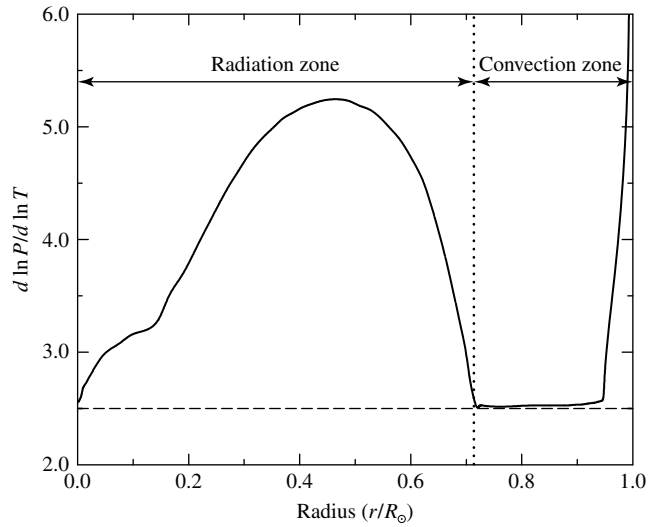
$$\left| \frac{dT}{dr} \right|_{\text{act}} > \left| \frac{dT}{dr} \right|_{\text{ad}},$$

where the “act” and “ad” subscripts designate the actual and adiabatic temperature gradients, respectively. Under the simplifying assumption of an ideal monatomic gas, this condition becomes (Eq. 10.95),

$$\frac{d \ln P}{d \ln T} < 2.5.$$

$d \ln P / d \ln T$  is plotted versus  $r/R_\odot$  in Fig. 11.7. As can be seen, the Sun is purely radiative below  $r/R_\odot = 0.714$  and becomes convective above that point. Physically this occurs because the opacity in the outer portion of the Sun becomes large enough to inhibit the transport of energy by radiation; recall that the radiative temperature gradient is proportional to the opacity (see Eq. 10.68). When the temperature gradient becomes too large, convection becomes the more efficient means of energy transport. Throughout most of the region of convective energy transport,  $d \ln P / d \ln T \simeq 2.5$ , which is characteristic of the nearly adiabatic temperature gradient of most convection zones. The rapid rise in  $d \ln P / d \ln T$





**FIGURE 11.7** The convection condition  $d \ln P / d \ln T$  plotted versus  $r/R_{\odot}$ . The dashed horizontal line represents the boundary between adiabatic convection and radiation for an ideal monatomic gas. The onset of convection does not exactly agree with the ideal adiabatic case because of the incorporation of a sophisticated equation of state and a more detailed treatment of convection physics. The rapid rise in  $d \ln P / d \ln T$  near the surface is associated with the highly superadiabatic nature of convection in that region (i.e., the adiabatic approximation that convection occurs when  $d \ln P / d \ln T < 2.5$  is invalid near the surface of the Sun). [ $d \ln P / d \ln T$  was computed using data from Bahcall, Pinsonneault, and Basu, *Ap. J.*, 555, 990, 2001. The data for the zones above  $0.95 R_{\odot}$  are from Cox, Arthur N. (editor), *Allen's Astrophysical Quantities*, Fourth Edition, AIP Press, New York, 2000.]

above  $0.95 R_{\odot}$  is due to the significant departure of the actual temperature gradient from an adiabatic one. In this case convection must be described by a more detailed treatment, such as the mixing-length theory discussed at the end of Section 10.4.

Notice that  $d \ln P / d \ln T$  also decreases to almost 2.5 at the center of the Sun. Although the Sun remains purely radiative in the center, the large amount of energy that must be transported outward pushes the temperature gradient in the direction of becoming superadiabatic. We will see in Chapter 13 that stars only slightly more massive than the Sun are convective in their centers because of the stronger temperature dependence of the CNO cycle as compared to the pp chain.

Clearly, an enormous amount of information is available regarding the solar interior, as derived from the direct and careful application of the stellar structure equations and the fundamental physical principles described in the last three chapters. A very complete and reasonable model of the Sun can be produced that is consistent with evolutionary timescales and fits the global characteristics of the star, specifically its mass, luminosity, radius, effective temperature, and surface composition; precise measurements of oscillation frequencies (see Chapter 14); and, as we will see in the next section, its observed surface convection zone.

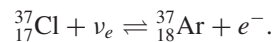
One aspect of the observed Sun that is not yet fully consistent with the current solar model is the abundance of lithium. The observed lithium abundance at the Sun's surface

is actually somewhat less than expected and may imply some need for adjustments in the model through refined treatments of convection, rotation, and/or mass loss. The lithium problem will be discussed further in Chapter 13.

### The Solar Neutrino Problem: A Detective Story Solved

Another significant discrepancy had existed between observations and the solar model for several decades, the resolution of which led to an important new understanding of fundamental physics. The **solar neutrino problem** was first noticed when Raymond Davis began measuring the neutrino flux from the Sun in 1970 using a detector located almost one mile below ground in the Homestake Gold Mine in Lead, South Dakota (Fig. 11.8). Because of the very low cross section of neutrino interactions with other matter, neutrinos can easily travel completely through Earth while other particles originating from space cannot. As a result, the underground detector was assured of measuring what it was designed to measure—neutrinos created eight minutes earlier in the solar core.

The Davis neutrino detector contained 615,000 kg of cleaning fluid,  $\text{C}_2\text{Cl}_4$  (tetrachlorethylene) in a volume of 377,000 liters (100,000 gallons). One isotope of chlorine ( $^{37}_{17}\text{Cl}$ ) is capable of interacting with neutrinos of sufficient energy to produce a radioactive isotope of argon that has a half-life of 35 days,

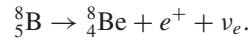


The threshold energy for this reaction, 0.814 MeV, is less than the energies of the neutrinos produced in every step of the pp chain except the crucial first one,  $^1_1\text{H} + ^1_1\text{H} \rightarrow ^2_1\text{H} + e^+ + \nu_e$  (recall the reaction sequences depicted in Fig. 10.8). However, the reaction that accounted



**FIGURE 11.8** Raymond Davis's solar neutrino detector. The tank was located 1478 m (4850 ft) below ground in the Homestake Gold Mine in Lead, South Dakota, and was filled with 615,000 kg of  $\text{C}_2\text{Cl}_4$  in a volume of 377,000 liters (100,000 gallons). (Courtesy of Brookhaven National Laboratory.)

for 77% of the neutrinos detected in the Davis experiment is the decay of  ${}^8_5\text{B}$  in the PP III chain,



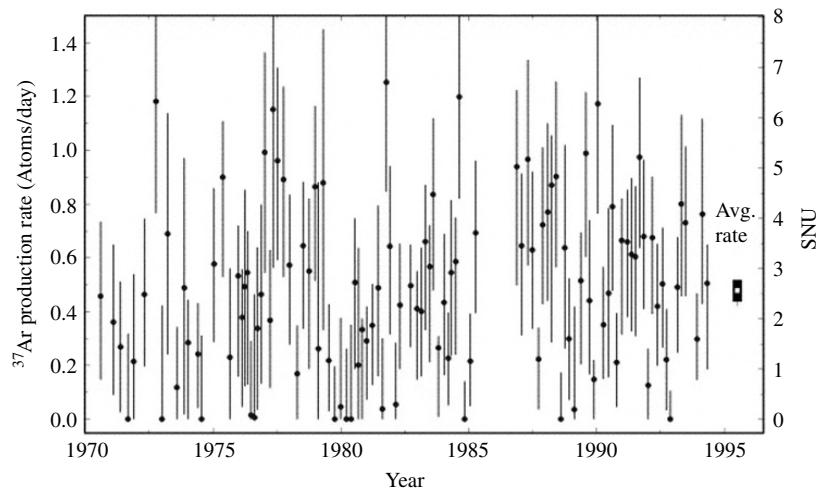
Unfortunately, this reaction is very rare, producing only for one pp chain termination in 5000.

John Bahcall (1935–2005), a colleague of Davis, was able to compute the anticipated rate at which solar neutrinos should have been detected by the chlorine experiment (the capture rate). The complex calculation was based on the rate of neutrino production by  ${}^8_5\text{B}$  decay in the PP III chain as computed from the solar model, combined with the probability that a solar neutrino will interact with a chlorine atom in the Homestake experiment.

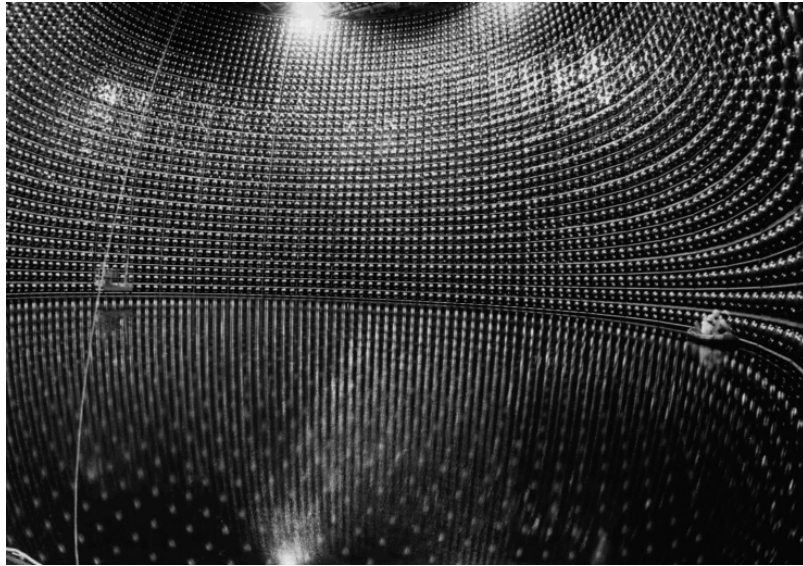
Once every few months Davis and his collaborators carefully purged the accumulated argon from the tank and determined the number of argon atoms produced. The capture rate was measured in terms of the **solar neutrino unit**, or SNU ( $1 \text{ SNU} \equiv 10^{-36}$  reactions per target atom per second). With approximately  $2.2 \times 10^{30}$  atoms of  ${}^{37}_{17}\text{Cl}$  atoms in the tank, if only one argon atom was produced each day, this rate would have corresponded to 5.35 SNU.

Results of 108 extractions from the Davis experiment between 1970 and 1994 are shown in Fig. 11.9. Bahcall predicted that the experiment should have yielded a capture rate of 7.9 SNU while the actual data gave an average of  $2.56 \pm 0.16$  SNU; only one argon atom was produced every two days in that 100,000 gallon tank!

Other neutrino experiments, fundamentally different from the  ${}^{37}_{17}\text{Cl}$  experiment, have confirmed the discrepancy between the prediction of the solar model and observed neutrino counts. Japan's underground Super-Kamiokande observatory (Fig. 11.10) detects the

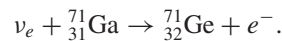


**FIGURE 11.9** Results of the Davis solar neutrino experiment from 1970 to 1994. The uncertainties in the experimental data are shown by vertical error bars associated with each run. The predicted solar neutrino capture rate for the  ${}^{37}_{17}\text{Cl}$  detector was 7.9 SNU based on solar models without neutrino oscillations. (Figure adapted from Cleveland, et al., *Ap. J.*, 496, 505, 1998.)



**FIGURE 11.10** Super-Kamiokande neutrino observatory in Japan contains  $4.5 \times 10^7$  kg (50,000 tons) of pure water. As neutrinos pass through the water, they scatter electrons at speeds greater than the speed of light through water. The pale blue Cerenkov light that is produced is detected by the 11,200 inwardly-directed photomultiplier tubes, signaling the presence of the passing neutrino. [Photo courtesy of Kamioka Observatory, ICRR (Institute for Cosmic Ray Research), The University of Tokyo.]

**Cerenkov light** that is produced when neutrinos scatter electrons, causing the electrons to move at speeds greater than the speed of light in water.<sup>6</sup> The number of neutrinos detected by Super-Kamiokande (and Kamiokande II before it) are less than half the number expected from solar models. The Soviet–American Gallium Experiment (SAGE), located at the Baksan Neutrino Laboratory (inside a mountain in the Caucasus), and GALLEX (at the Gran Sasso underground laboratory in Italy) measure the low-energy pp chain neutrinos that dominate the Sun’s neutrino flux. SAGE and GALLEX make their detections via a reaction that converts gallium into germanium,



After considering the expected number of background counts from sources other than the Sun, both experiments also confirm the deficit of neutrinos first established by the Davis detector.

The search for a theoretical resolution to the solar neutrino problem considered two general approaches: Either some fundamental physical process operating in the solar model is incorrect, or something happens to the neutrinos on their way from the Sun’s core to Earth. The first of these possibilities inspired an intense reexamination of a host of features of the

<sup>6</sup>Note that this *does not* violate Einstein’s special theory of relativity since the special theory applies to the speed of light in a vacuum. The speed of light in any other medium is always less than the speed of light in a vacuum.

solar model, including nuclear reaction rates, the opacity of stellar material, the evolution of the Sun up to its present state, variations in the composition of the solar interior, and several exotic suggestions (including *dark matter* in the Sun's core). However, none of these suggested solutions was able to satisfy all of the observational constraints simultaneously, particularly neutrino counts and solar oscillation frequencies.

An elegant solution to the solar neutrino problem proposed that the solar model is essentially correct but that the neutrinos produced in the Sun's core actually change before they reach Earth. The **Mikheyev–Smirnov–Wolfenstein** (or **MSW**) **effect** involves the transformation of neutrinos from one type to another. This idea is an extension of the *electroweak theory* of particle physics that combines the electromagnetic theory with the theory of weak interactions governing some types of radioactive decay. The neutrinos produced in the various branches of the pp chain are all electron neutrinos ( $\nu_e$ ); however, two other *flavors* of neutrinos also exist—the muon neutrino ( $\nu_\mu$ ) and the tau neutrino ( $\nu_\tau$ ). The MSW effect suggests that neutrinos oscillate among flavors, being electron neutrinos, muon neutrinos, and/or tau neutrinos during their passage through the Sun. The neutrino oscillations are caused by interactions with electrons as the neutrinos travel toward the surface. Because the chlorine (Davis), water (Kamiokande and Super-Kamiokande), and gallium detectors (SAGE and GALLEX) have different threshold energies and they are sensitive only to the electron neutrino, their results were determined to be consistent with the MSW theory.

One testable consequence of the MSW effect is that if neutrinos oscillate between flavors, they must necessarily have mass. This is because a change of neutrino flavor can occur only between neutrinos having different masses. The required mass difference needed for the MSW solution to the solar neutrino problem is much less than the current experimentally established upper limit on the mass of the electron neutrino of approximately 2.2 eV. Even though the standard electroweak theory does not predict masses for the neutrinos, many reasonable extensions of this theory do allow for masses in the right range. These extended theories, known as **grand unified theories** (GUTs), are currently the focus of intense research by high-energy (particle) physicists.

Confirmation of neutrino oscillations came in 1998 when Super-Kamiokande was used to detect *atmospheric neutrinos* that are produced when high-energy cosmic rays (charged particles from space) collide with Earth's upper atmosphere. Cosmic rays are capable of creating both electron and muon neutrinos, but not tau neutrinos. The Super-Kamiokande group was able to determine that the number of muon neutrinos traveling upward after having traversed the diameter of Earth was significantly reduced relative to the number traveling downward. The difference in numbers is in excellent agreement with the theory of neutrino mixing (neutrinos oscillating among the three flavors), demonstrating for the first time that neutrinos are not massless particles.

Thus, after several decades of study, the solar neutrino problem was resolved by a profound advance in our understanding of particle physics and the nature of the fundamental forces. As a result of their contributions to this important scientific detective story, Raymond Davis and Masatoshi Koshiba, director of the Kamiokande research group that confirmed the neutrino detections, were two of the recipients of the 2002 Nobel Prize in physics.<sup>7</sup>

<sup>7</sup>The third recipient of the 2002 Nobel Prize, Riccardo Giacconi, used a rocket experiment to detect X-rays in space. Giacconi later designed the Uhuru and Einstein X-ray observatories and also served as the first director of the Space Telescope Science Institute.



In 2004, John Bahcall wrote of the efforts to solve the solar neutrino problem:

I am astonished when I look back on what has been accomplished in the field of solar neutrino research over the past four decades. Working together, an international community of thousands of physicists, chemists, astronomers, and engineers has shown that counting radioactive atoms in a swimming pool full of cleaning fluid in a deep mine on Earth can tell us important things about the center of the Sun and about the properties of exotic fundamental particles called neutrinos. If I had not lived through the solar neutrino saga, I would not have believed it was possible.<sup>8</sup>

## 11.2 ■ THE SOLAR ATMOSPHERE

When we observe the Sun visually, it appears as though there is a very abrupt and clear edge to this hot, gaseous ball (Fig. 11.11). Of course, an actual “surface” does not exist; rather, what we are seeing is a region where the solar atmosphere is *optically thin* and photons originating from that level travel unimpeded through space. Even this region is not clearly defined, however, since some photons can always escape when the optical depth is somewhat greater than unity while others may be absorbed when the optical depth is less than unity, but the odds of a photon leaving the solar atmosphere diminish rapidly as the optical depth increases. Consequently, the Sun’s atmosphere changes from being optically thin to optically thick in only about 600 km. This relatively small distance (about 0.09% of the Sun’s radius) is what gives the “edge” of the Sun its sharp appearance.

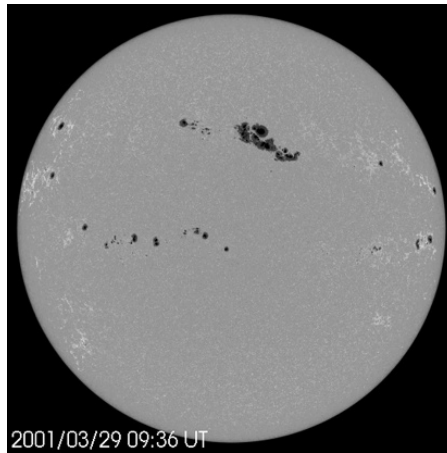
### The Photosphere

The region where the observed optical photons originate is known as the solar **photosphere**. Defining the base of the photosphere is somewhat arbitrary since some photons can originate from an optical depth significantly greater than unity. For instance, if 1% of the photons originating from a layer reach us, the optical depth would be approximately 4.5 at that level ( $e^{-4.5} \sim 0.01$ ); if 0.1% reach us, the optical depth would be about 6.9. Of course, since the opacity and optical depth are wavelength dependent, the base of the photosphere is also wavelength dependent if it is defined in terms of the optical depth. Given the arbitrariness of the definition, the base of the photosphere for the Sun is sometimes simply defined to be 100 km below the level where the optical depth at a wavelength of 500 nm is unity. At this depth,  $\tau_{500} \simeq 23.6$  and the temperature is approximately 9400 K.

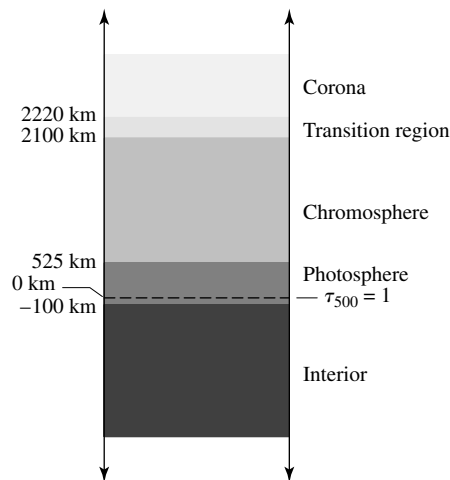
Moving upward through the solar photosphere, the temperature of the gas decreases from its base value to a minimum of 4400 K about 525 km above the  $\tau_{500} = 1$  level. It is this temperature minimum that defines the top of the photosphere. Above this point, the temperature begins to rise again. The approximate thicknesses of the various components of the Sun’s atmosphere to be discussed in this section are depicted in Fig. 11.12.

As was discussed in Section 9.4, on average the solar flux is emitted from an optical depth of  $\tau = 2/3$  (the Eddington approximation). This leads to the identification of the effective temperature with the temperature of the gas at this depth, or  $T_e = T_{\tau=2/3} = 5777$  K.

<sup>8</sup>“Solving the Mystery of the Missing Neutrinos,” John N. Bahcall (2004), Nobel e-Museum, <http://nobelprize.org/physics/articles/bahcall/>.



**FIGURE 11.11** The solar disk appears sharp because of the rapid increase in optical depth with distance through the photosphere. Sunspots are visible on the surface of the disk in this image taken by SOHO/MDI on March 29, 2001. [SOHO (ESA & NASA)]



**FIGURE 11.12** The thicknesses of the components of the Sun's atmosphere.

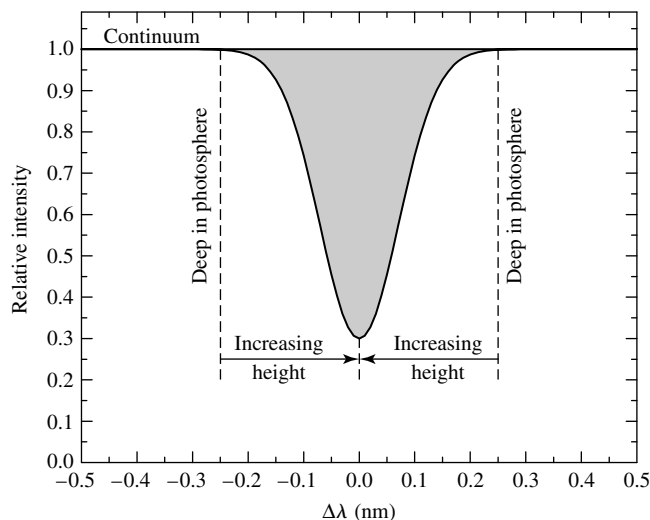
Recall that the Sun radiates predominantly as a blackbody in the visible and infrared portions of the spectrum (note the relatively smooth features of the solar spectrum depicted in Fig. 9.5). This observation suggests that there exists a source of opacity that is basically continuous across wavelength. The continuum opacity is due in part to the presence of the  $\text{H}^-$  ions in the photosphere (recall the discussion on page 248).

Using the Saha equation (Eq. 8.8), we can determine the ratio of the number of  $\text{H}^-$  ions to neutral hydrogen atoms. It is left as an exercise to show that in the Sun's photosphere,

only about one in  $10^7$  hydrogen atoms actually forms an  $\text{H}^-$  ion. The importance of  $\text{H}^-$  in the Sun is due to the fact that even though the abundance of the ion is quite low, neutral hydrogen is not capable of contributing significantly to the continuum.

Of course, optical depth is a function not only of the distance that a photon must travel to the surface of the Sun, but also of the wavelength-dependent opacity of the solar material (Eq. 9.17). Consequently, photons can originate from or be absorbed at different physical depths in the atmosphere, depending upon their wavelengths. Since a spectral line is not infinitesimally thin, but actually covers a range of wavelengths (recall Section 9.5), even different parts of the same line are formed at different levels of the atmosphere. Thus solar observations with high-wavelength resolution can be used to probe the atmosphere at various depths, providing a wealth of information about its structure.

Absorption lines, including Fraunhofer lines, are produced in the photosphere (see Section 5.1). According to Kirchhoff's laws, the absorption lines must be produced where the gas is cooler than the bulk of the continuum-forming region. Line formation must also occur between the observer and the region where much of the continuum is produced. In reality, the Fraunhofer lines are formed in the same layers where  $\text{H}^-$  produces the continuum. However, the darkest part of the line (its center) originates from regions higher in the photosphere, where the gas is cooler. This is because the opacity is greatest in the center of the line, making it more difficult to see deeper into the photosphere. Moving away from the central wavelength toward the wing of the line implies that absorption is occurring at progressively deeper levels. At wavelengths sufficiently far from the central peak, the edge of the line merges with the continuum being produced at the base of the photosphere. This effect is illustrated in Fig. 11.13 (recall the discussion at the beginning of Section 9.5).



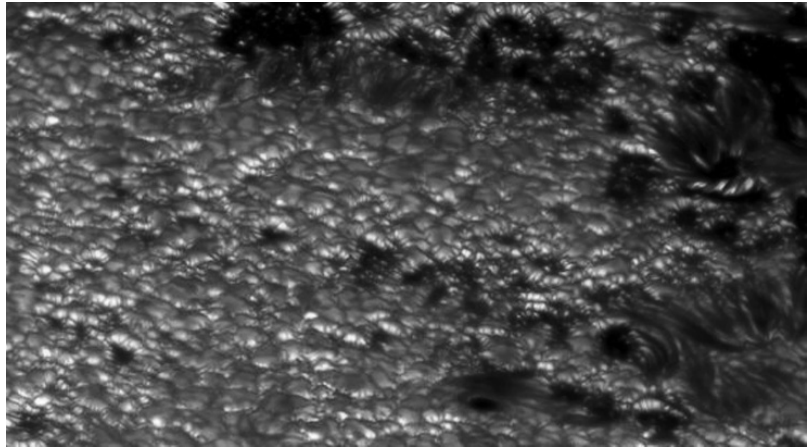
**FIGURE 11.13** The relationship between absorption line strength and depth in the photosphere for a typical spectral line. The wings of the line are formed deeper in the photosphere than is the center of the line.



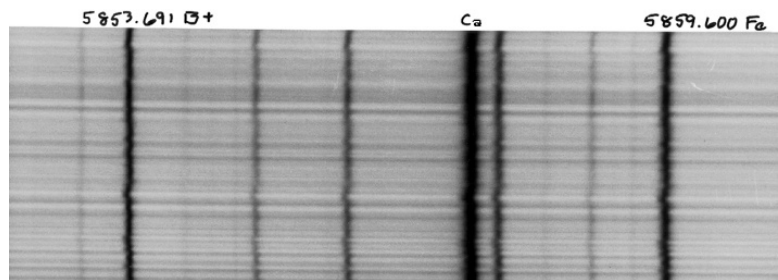
### Solar Granulation

When the base of the photosphere is observed (see Fig. 11.14), it appears as a patchwork of bright and dark regions that are constantly changing, with individual regions appearing and then disappearing. With a spatial extent of roughly 700 km, the characteristic lifetime for one of these regions is five to ten minutes. This patchwork structure is known as **granulation** and is the top of the convection zone protruding into the base of the photosphere.

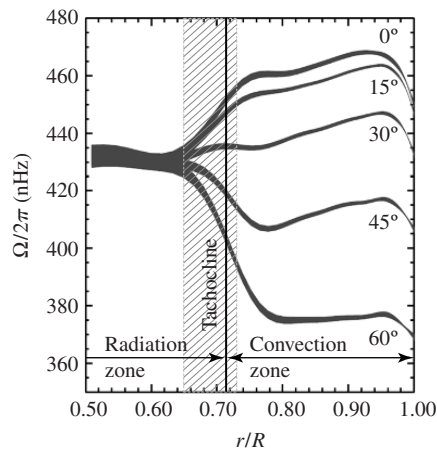
Figure 11.15 shows a high-resolution spectrum of solar granulation spanning a number of convection cells. The appearance of wiggles in the absorption lines occurs because some parts of the region are Doppler blueshifted while others are redshifted. Using Eq. (4.39), we



**FIGURE 11.14** Granulation at the base of the photosphere is due to the rising and falling gas bubbles produced by the underlying convection zone. (This three-dimensional image is from the Swedish 1-m Solar Telescope, operated on the island of La Palma by the Institute for Solar Physics of the Royal Swedish Academy of Sciences in the Spanish Observatorio del Roque de los Muchachos of the Instituto de Astrofísica de Canarias.)



**FIGURE 11.15** A spectrum of a portion of the photospheric granulation showing absorption lines that indicate the presence of radial motions. Wiggles to the left are toward shorter wavelengths and are blueshifted while wiggles to the right are redshifted. The wavelengths shown at the top of the image are given in angstroms. (Courtesy of W. Livingston and the National Optical Astronomy Observatories.)



**FIGURE 11.16** The rotation period of the Sun varies with latitude and depth.  $\Omega$ , the angular frequency, has units of radians per second. (Adapted from a figure courtesy of NSF's National Solar Observatory.)

find that radial velocities of  $0.4 \text{ km s}^{-1}$  are common; brighter regions produce the blueshifted sections of the lines while darker regions produce the redshifted sections. Thus the bright cells are the vertically rising hot convective bubbles carrying energy from the solar interior. When those bubbles reach the optically thin photosphere, the energy is released via photons and the resulting cooler, darker gas sinks back into the interior. The lifetime of a typical granule is the amount of time needed for a convective eddy to rise and fall the distance of one mixing length. Solar granulation provides us with a visual verification of the results of the stellar structure equations applied to our Sun.

### Differential Rotation

Photospheric absorption lines may also be used to measure the rotation rate of the Sun. By measuring Doppler shifts at the solar limb, we find that the Sun rotates *differentially* (i.e., the rate of rotation depends on the latitude being observed). At the equator the rotation period is approximately 25 days, increasing to 36 days at the poles.

Observations of solar oscillations have revealed that the Sun's rotation also varies with radius; see Fig. 11.16. Near the base of the convection zone, the differing rotation rates with latitude converge in a region known as the **tachocline**. The strong shear that is set up in this region is believed to result in electric currents in the highly conducting plasma, which in turn generate the Sun's magnetic field. Thus the tachocline is probably the source of the Sun's magnetic field. (The complex manifestations of the Sun's dynamic magnetic field will be discussed extensively in Section 11.3.)

### The Chromosphere

The **chromosphere**, with an intensity that is only about  $10^{-4}$  of the value for the photosphere, is that portion of the solar atmosphere that lies just above the photosphere and

extends upward for approximately 1600 km (2100 km above  $\tau_{500} = 1$ ). Analysis of the light produced in the chromosphere indicates that the gas density drops by more than a factor of  $10^4$  and that the temperature begins to *increase* with increasing altitude, from 4400 K to about 10,000 K.

Reference to the Boltzmann and Saha equations (8.6 and 8.8, respectively) shows that lines that are not produced at the lower temperatures and higher densities of the photosphere can form in the environment of the chromosphere. For instance, along with the hydrogen Balmer lines, the lines of He II, Fe II, Si II, Cr II, and Ca II (in particular, the Ca II H and K lines, 396.8 nm and 393.3 nm, respectively) can appear in the spectrum.

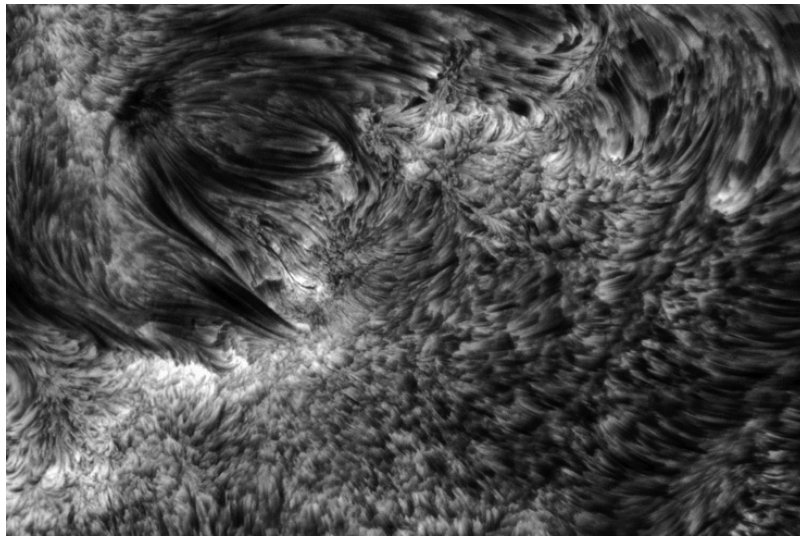
Although certain Fraunhofer lines appear as absorption lines in the visible and near ultraviolet portions of the spectrum, others begin to appear as emission lines at shorter (and much longer) wavelengths. Again Kirchhoff's laws offer an explanation, suggesting that a hot, low-density gas must be responsible. Because the interior of the Sun is optically thick below the base of the photosphere, the area of emission line production must occur elsewhere. With the peak of the blackbody spectrum near 500 nm, the strength of the continuum decreases rapidly at shorter and longer wavelengths (Fig. 3.8). As a result, emission lines produced outside of the visible portion of the spectrum are not overwhelmed by the blackbody radiation.

Visible wavelength emission lines are not normally seen against the bright solar disk, but they can be observed near the limb of the Sun for a few seconds at the beginning and end of a total eclipse of the Sun; this phenomenon is referred to as a **flash spectrum**. During this period, the portion of the Sun that is still visible takes on a reddish hue because of the dominance of the Balmer H $\alpha$  emission line, a line that is normally observed only as an absorption line in the Sun's atmosphere.

Using filters that restrict observations to the wavelengths of the emission lines produced in the chromosphere (particularly H $\alpha$ ), it is possible to see a great deal of structure in this portion of the atmosphere. **Supergranulation** becomes evident on scales of 30,000 km, showing the continued effects of the underlying convection zone. Doppler studies again reveal convective velocities on the order of  $0.4 \text{ km s}^{-1}$ , with gas rising in the centers of the supergranules and sinking at their edges. Also present are vertical filaments of gas, known as **spicules**, extending upward from the chromosphere for 10,000 km (Fig. 11.17). An individual spicule may have a lifetime of only 15 minutes, but at any given moment spicules cover several percent of the surface of the Sun. Doppler studies show that mass motions are present in spicules, with material moving outward at approximately  $15 \text{ km s}^{-1}$ .

### The Transition Region

Above the chromosphere, the temperature rises very rapidly within approximately 100 km (see Fig. 11.18), reaching more than  $10^5 \text{ K}$  before the temperature gradient flattens somewhat. The temperature then continues to rise more slowly, eventually exceeding  $10^6 \text{ K}$ . This **transition region** may be selectively observed at various altitudes in the ultraviolet and extreme ultraviolet parts of the electromagnetic spectrum. For instance, the 121.6-nm Lyman-alpha (Ly $\alpha$ ) emission line of hydrogen ( $n = 2 \rightarrow n = 1$ ) is produced at the top of the chromosphere at 20,000 K, the C III 97.7-nm line originates at a level where the temperature is 90,000 K, the 103.2-nm line of O VI occurs at 300,000 K, and Mg X creates a



**FIGURE 11.17** Spicules in the chromosphere of the Sun. In addition, small sunspots are visible in the upper left quadrant of the image, and brighter areas known as plage regions are also visible. The observations were made using the  $H\alpha$  emission line. Features as small as 130 km are evident in this image. (Courtesy of the Royal Swedish Academy of Sciences.)

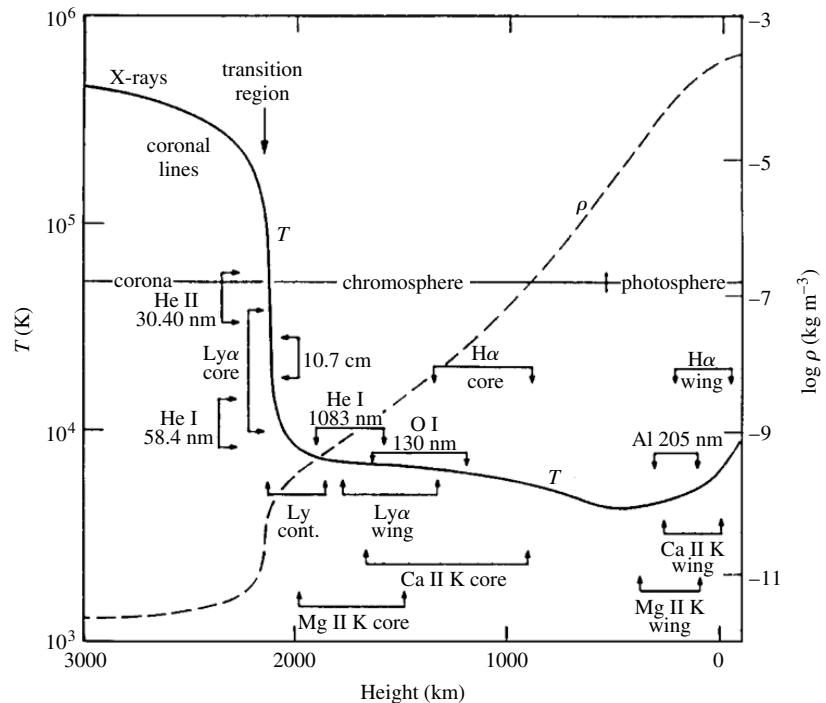
62.5-nm line at  $1.4 \times 10^6$  K. Figure 11.19 shows images of the Sun at various wavelengths and heights above the base of the photosphere.

### The Corona

When the Moon fully occults the photosphere during a total solar eclipse, the radiation from the faint **corona** becomes visible (Fig. 11.20). The corona, located above the transition region, extends out into space without a well-defined outer boundary and has an energy output that is nearly  $10^6$  times less intense than that of the photosphere. The number density of particles at the base of the corona is typically  $10^{15}$  particles  $\text{m}^{-3}$ , whereas in the vicinity of Earth, the number density of particles originating from the Sun (solar wind particles) have a characteristic value of  $10^7$  particles  $\text{m}^{-3}$  (this can be compared with  $10^{25}$  particles  $\text{m}^{-3}$  at sea level in Earth's atmosphere). Because the density of the corona is so low, it is essentially transparent to most electromagnetic radiation (except long radio wavelengths) and is not in local thermodynamic equilibrium (LTE). For gases that are not in LTE, a unique temperature is not strictly definable (see Section 9.2). However, the temperatures obtained by considering thermal motions, ionization levels, and radio emissions do give reasonably consistent results. For instance, the presence of Fe XIV lines indicates temperatures in excess of  $2 \times 10^6$  K, as do line widths produced by thermal Doppler broadening.

Based on the radiation coming from the corona, three distinct structural components can be identified:

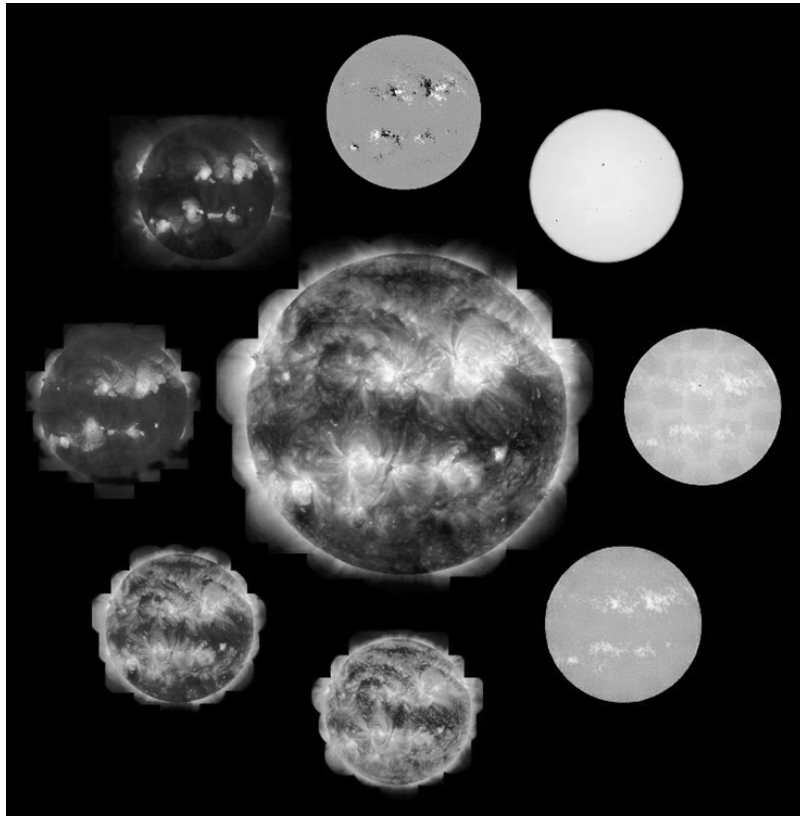
- The **K corona** (from *Kontinuierlich*, the German word for “continuous”) produces the continuous white light emission that results from photospheric radiation scattered



**FIGURE 11.18** Logarithmic plots of the temperature structure (solid line) and mass density structure (dashed line) of the upper atmosphere of the Sun. The altitudes observed at various wavelengths are also depicted. [Figure adapted from Avrett, in *Encyclopedia of Astronomy and Astrophysics*, Paul Murdin (ed.), Institute of Physics Publishing, Bristol, 2001, page 2480.]

by free electrons. Contributions to the coronal light due to the K corona primarily occur between 1 and  $2.3 R_{\odot}$  from the center of the Sun. The spectral lines evident in the photosphere are essentially blended by the large Doppler shifts that are caused by the high thermal velocities of the electrons.

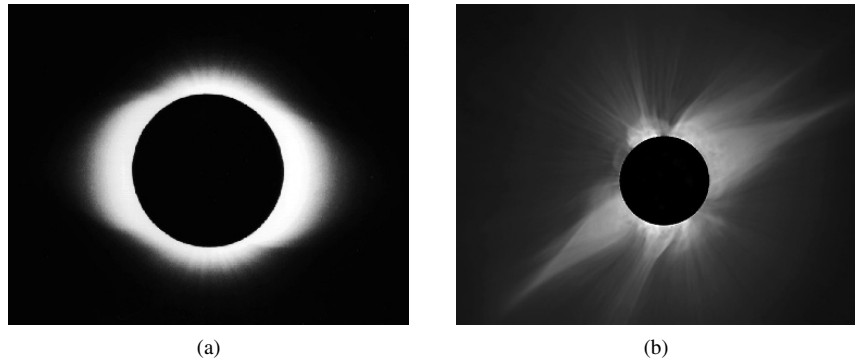
- The **F corona** (for Fraunhofer) comes from the scattering of photospheric light by dust grains that are located beyond  $2.3 R_{\odot}$ . Because dust grains are much more massive and slower than electrons, Doppler broadening is minimal and the Fraunhofer lines are still detectable. The F corona actually merges with the **zodiacal light**, the faint glow found along the ecliptic that is a reflection of the Sun's light from interplanetary dust.
- The **E corona** is the source of the emission lines that are produced by the highly ionized atoms located throughout the corona; the E corona overlaps the K and F coronas. Since the temperatures are extremely high in the corona, the exponential term in the Saha equation encourages ionization because thermal energies are comparable to ionization potentials. The very low number densities also encourage ionization since the chance of recombination is greatly reduced.



**FIGURE 11.19** Visible features of the Sun at various wavelengths. The central image is a three-color composite of the corona obtained by TRACE at 17.1 nm, 19.5 nm, and 28.4 nm. Clockwise starting from the top are a SOHO/MDI magnetic map, white light, TRACE 170 nm continuum, TRACE Ly $\alpha$ , TRACE 17.1 nm, TRACE 19.5 nm, TRACE 28.4 nm, and a Yohkoh/SXT X-ray image. [The Transition Region and Coronal Explorer, TRACE, is a mission of the Stanford-Lockheed Institute for Space Research (a joint program of the Lockheed-Martin Advanced Technology Center's Solar and Astrophysics Laboratory and Stanford's Solar Observatories Group) and part of the NASA Small Explorer program.]

The low number densities allow forbidden transitions to occur, producing spectral lines that are generally seen only in astrophysical environments where gases are extremely thin (recall the discussion of selection rules in Section 5.4). Forbidden transitions occur from atomic energy levels that are **metastable**; electrons do not readily make transitions from metastable states to lower energy states without assistance. Whereas allowed transitions occur on timescales on the order of  $10^{-8}$  s, spontaneous forbidden transitions may require one second or longer. In gases at higher densities, electrons are able to escape from metastable states through collisions with other atoms or ions, but in the corona these collisions are rare. Consequently, given enough time, some electrons will be able to make spontaneous transitions from metastable states to lower energy states, accompanied by the emission of photons.

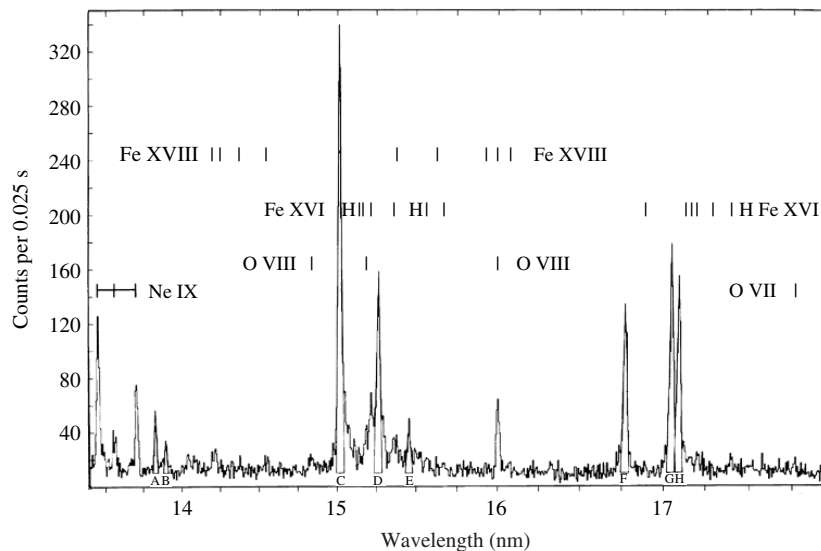




**FIGURE 11.20** (a) The quiet solar corona seen during a total solar eclipse in 1954. The shape of the corona is elongated along the Sun's equator. (Courtesy of J. D. R. Bahng and K. L. Hallam.) (b) The active corona tends to have a very complex structure. This image of the July 11, 1991, eclipse is a composite of five photographs that was processed electronically. (Courtesy of S. Albers.)

Since the blackbody continuum emission from the Sun decreases like  $\lambda^{-4}$  for sufficiently long wavelengths (see Eq. 3.20), the amount of photospheric radio emission is negligible. The solar corona, however, is a source of radio-wavelength radiation that is not associated with the blackbody continuum. Some radio emission arises from free-free transitions of electrons that pass near atoms and ions. During these close encounters, photons may be emitted as the electrons' energies are decreased slightly. From the conservation of energy, the greater the change in the energy of an electron, the more energetic the resulting photon and the shorter its wavelength. Clearly, the closer an electron comes to an ion, the more likely it is that the electron's energy will change appreciably. Since more frequent and closer encounters are expected if the number density is larger, shorter-wavelength radio emissions should be observed nearer the Sun. Radio wavelengths of 1 to 20 cm are observed from the chromosphere through the lower corona, while longer wavelength radiation originates from the outer corona. It is important to note that synchrotron radiation by relativistic electrons also contributes to the observed radio emission from the solar corona (recall the discussion of the headlight effect in Example 4.3.3).

Photospheric emissions are negligible in the X-ray wavelength range as well. In this case the blackbody continuum decreases very rapidly, dropping off like  $\lambda^{-5}e^{-hc/\lambda kT}$ . Consequently, any emission in X-ray wavelengths from the corona will completely overwhelm the output from the photosphere. In fact, because of the high temperatures of the corona, its X-ray spectrum is very rich in emission lines. This is due to the high degree of ionization that exists for all of the elements present, together with the ability of the corona to excite a large number of atomic transitions. Given the many electrons that are present in heavy elements such as iron and the vast number of available energy levels, each such element is capable of producing an extensive emission spectrum. Figure 11.21 shows a section of the X-ray emission spectrum of the solar corona. It displays a sample of the lines that are observed in one portion of the X-ray wavelength band, along with the ions responsible for their production.



**FIGURE 11.21** A section of the X-ray emission spectrum of the solar corona. (Figure adapted from Parkinson, *Astron. Astrophys.*, 24, 215, 1973.)

### Coronal Holes and the Solar Wind

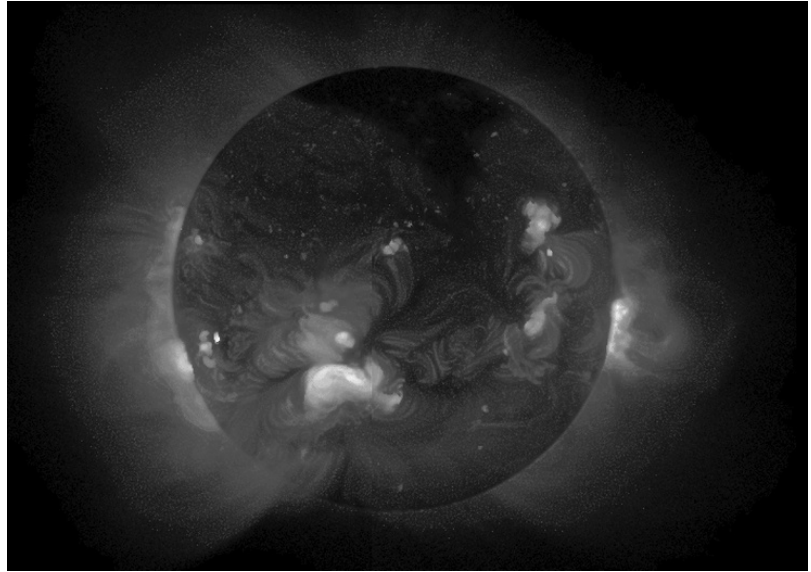
An image of the X-ray Sun is shown in Fig. 11.22. This fascinating picture indicates that X-ray emission is not uniform. Active (bright and hot) regions exist, along with darker, cooler regions known as **coronal holes**. Moreover, even in the coronal holes, localized bright spots of enhanced X-ray emission appear and disappear on a timescale of several hours. Smaller features are also apparent within the regions of generally bright X-ray emission.

The weaker X-ray emission coming from coronal holes is characteristic of the lower densities and temperatures that exist in those regions, as compared to the rest of the corona. The explanation for the existence of coronal holes is tied to the Sun's magnetic field and the generation of the **fast solar wind**, a continuous stream of ions and electrons escaping from the Sun and moving through interplanetary space at speeds of approximately  $750 \text{ km s}^{-1}$ . A gusty, **slow solar wind**, with speeds of roughly one-half those of the fast wind appears to be produced by streamers in the corona associated with closed magnetic fields.

Just like the magnetic field that is produced by a current loop, the magnetic field of the Sun is generally that of a dipole, at least on a global scale (Fig. 11.23). Although its value can differ significantly in localized regions (as we will see in the next section), the strength of the field is typically a few times  $10^{-4} \text{ T}$  near the surface.<sup>9</sup> Coronal holes correspond to those parts of the magnetic field where the field lines are *open*, while the X-ray bright regions are associated with *closed* field lines; open field lines extend out to great distances from the Sun, while closed lines form loops that return to the Sun.

<sup>9</sup>The magnetic field near the surface of Earth is approximately  $6 \times 10^{-5} \text{ T}$ .





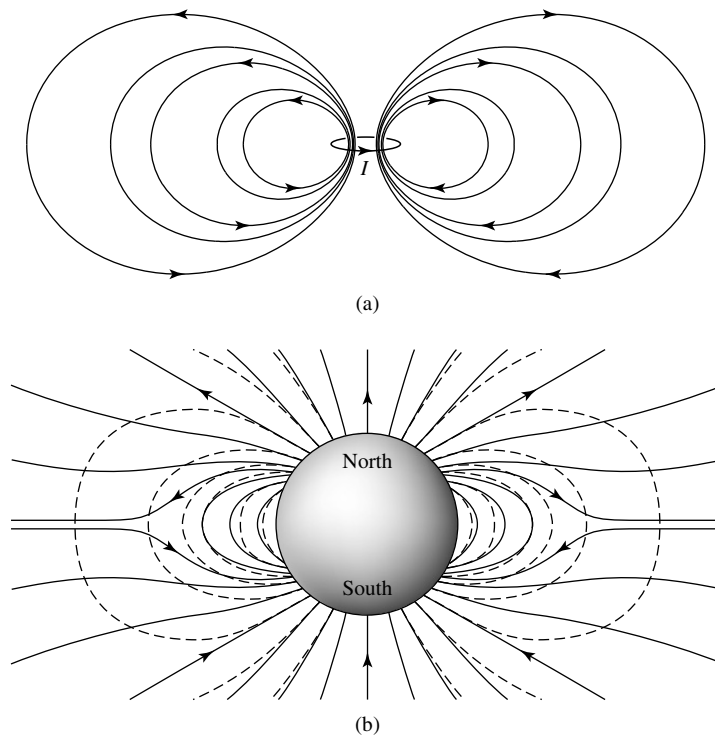
**FIGURE 11.22** An X-ray image of the Sun obtained by the Soft X-Ray Telescope on the Yohkoh Solar Observatory, May 8, 1992. Bright regions are hotter X-ray regions and darker regions are cooler. A dark coronal hole is evident at the top of the image. (From the Yohkoh mission of ISAS, Japan. The X-ray telescope was prepared by the Lockheed Palo Alto Research Laboratory, the National Astronomical Observatory of Japan, and the University of Tokyo with the support of NASA and ISAS.)

The Lorentz force equation,

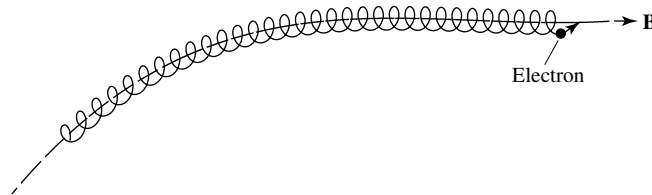
$$\mathbf{F} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (11.2)$$

describing the force exerted on a charged particle of velocity  $\mathbf{v}$  in an electric field  $\mathbf{E}$  and a magnetic field  $\mathbf{B}$  states that the force due to the magnetic field is always mutually perpendicular to both the direction of the velocity vector and the field (the cross product). Providing that electric fields are negligible, charged particles are forced to spiral around magnetic field lines and cannot actually cross them except by collisions (Fig. 11.24). This implies that closed magnetic field lines tend to trap charged particles, not allowing them to escape. In regions of open field lines, however, particles can actually follow the lines out away from the Sun. Consequently, the solar wind originates from the regions of open magnetic field lines, namely the coronal holes. The details observed in the X-ray-bright regions, as well as the localized bright spots in the coronal holes, are due to the higher densities of the electrons and ions that are trapped in large and small magnetic field loops.

The existence of ongoing mass loss from the Sun was deduced long before it was ever detected directly, as evidenced by the tails of comets. The tails are generally composed of two parts, a curved dust tail and a straight ion tail, both of which are always pointed away from the Sun (Fig. 11.25). The force exerted on dust grains by photons (radiation pressure)

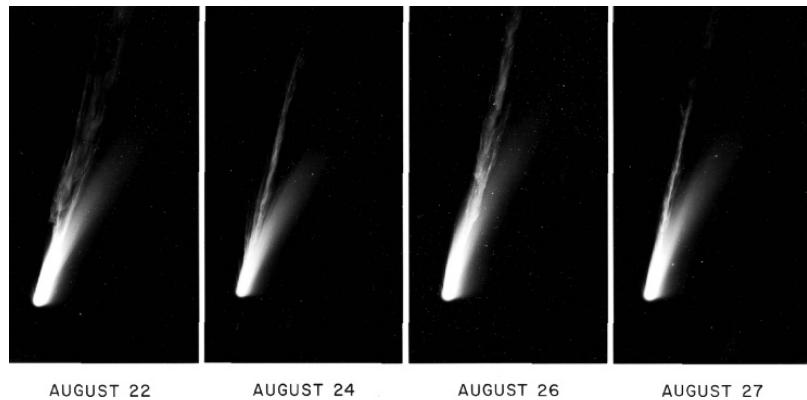


**FIGURE 11.23** (a) The characteristic dipole magnetic field of a current loop. (b) A generalized depiction of the global magnetic field of the Sun. The dashed lines show the field of a perfect magnetic dipole.



**FIGURE 11.24** A charged particle is forced to spiral around a magnetic field line because the Lorentz force is mutually perpendicular to both the velocity of the particle and the direction of the magnetic field.

is sufficient to push the dust tail back; the curvature of the tail is due to the different orbital speeds of the individual dust grains, which, according to Kepler's third law, are a function of their varying distances from the Sun. However, the ion tail cannot be explained by radiation pressure; the interaction between photons and the ions is not efficient enough. Rather, it is the electric force between the ions of the solar wind and the ions in the comet that counts for the direction of the ion tail. This interaction allows momentum to be transferred to the cometary ions, driving them straight away from the Sun.



**FIGURE 11.25** Comet Mrkos in 1957. The dust tail of a comet is curved and its ion tail is straight. (Courtesy of Palomar/Caltech.)



**FIGURE 11.26** Aurora australis seen over the South Pole. (NASA)

The **aurora borealis** and the **aurora australis** (the northern and southern lights, respectively) are also products of the solar wind (see Fig. 11.26). As the ions from the Sun interact with Earth's magnetic field, they become trapped in it. Bouncing back and forth between the north magnetic pole and the south magnetic pole, these ions form the **Van Allen radiation belts**. Ions that are sufficiently energetic will collide with the atoms in Earth's upper atmosphere near the magnetic poles, causing the atmospheric atoms to become excited or ionized. The resulting de-excitations or recombinations emit the photons that produce the spectacular light displays observed from high northern and southern latitudes.

Using rockets and satellites, characteristics of the two solar winds can be measured as they pass near Earth. In addition, the Ulysses spacecraft, placed in a polar orbit around the Sun, was able to detect the wind well out of the plane of Earth's orbit. At a distance of 1 AU from the Sun, the solar wind velocity ranges from approximately  $200 \text{ km s}^{-1}$  to  $750 \text{ km s}^{-1}$ , with a typical density of  $7 \times 10^6 \text{ ions m}^{-3}$  and characteristic kinetic temperatures of  $4 \times 10^4 \text{ K}$  for protons and  $10^5 \text{ K}$  for electrons. Although the winds are composed primarily of protons and electrons, heavier ions are present as well.

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**Example 11.2.1.** The mass loss rate of the Sun may be estimated from the data given above. We know that all of the mass leaving the Sun must also pass through a sphere of radius 1 AU centered on the Sun; otherwise it would collect at some location in space. If we further assume (for simplicity) that the mass loss rate is spherically symmetric, then the amount of mass crossing a spherical surface of radius  $r$  in an amount of time  $t$  is just the mass density of the gas multiplied by the volume of the shell of gas that can travel across the sphere during that time interval, or

$$dM = \rho dV = (nm_H)(4\pi r^2 v dt),$$

where  $n$  is the number density of ions (mostly hydrogen),  $m_H$  is approximately the mass of a hydrogen ion,  $v$  is the ion velocity, and  $dV = A dr \simeq 4\pi r^2 v dt$  is the volume of a shell that crosses a spherical surface in an amount of time  $dt$ . Dividing both sides by  $dt$ , we obtain the mass loss rate,

$$\frac{dM}{dt} = 4\pi r^2 nm_H v = 4\pi r^2 \rho v. \quad (11.3)$$

By convention, stellar mass loss rates are generally given in *solar masses per year* and symbolized by  $\dot{M} \equiv dM/dt$ . Using  $v = 500 \text{ km s}^{-1}$ ,  $r = 1 \text{ AU}$ , and  $n = 7 \times 10^6 \text{ protons m}^{-3}$ , we find that

$$\dot{M}_\odot \simeq 3 \times 10^{-14} M_\odot \text{ yr}^{-1}.$$

At this rate it would require more than  $10^{13} \text{ yr}$  before the entire mass of the Sun is dissipated. However, the interior structure of the Sun is changing much more rapidly than this, so the effect of the present-day solar wind on the evolution of the Sun is minimal.

As an interesting aside, in 1992 both *Voyagers I* and *II* detected radio noise at frequencies of 1.8 to 3.5 kHz originating from the outer reaches of the Solar System. It is believed that the noise is produced where particles from the solar wind collide with the interstellar medium, producing a termination shock. (The interstellar medium is the dust and gas located between the stars; see Section 12.1). The 1992 observations represented the first detection of the **heliopause**, the outer limit of the Sun's electromagnetic influence. In 2005, when Voyager I was about 95 AU from Earth and traveling at 3.6 AU per year, it passed through the termination shock into the region known as the **heliosheath**. The strongest evidence that Voyager I did in fact cross the termination shock comes from the measurement of a sudden significant increase in the strength of the magnetic field that is carried by the solar wind. This increased magnetic field strength is due to the slowing of the solar wind particles and the resulting increase in particle density.

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### The Parker Wind Model

We now consider how the expansion of the solar corona produces the solar wind. This is a result of the corona's high temperature, together with the high thermal conductivity of the ionized gas, referred to as a **plasma**. The ability of the plasma to conduct heat implies that the corona is almost isothermal (recall Fig. 11.18).

In 1958 Eugene Parker developed an approximately isothermal model of the solar wind that has been successful in describing many of its basic features. To see why the solar wind is inevitable, begin by considering the condition of hydrostatic equilibrium, Eq. (10.6). If the mass of the corona is insignificant compared to the total mass of the Sun, then  $M_r \simeq M_\odot$  in that region and the hydrostatic equilibrium equation becomes

$$\frac{dP}{dr} = -\frac{GM_\odot\rho}{r^2}. \quad (11.4)$$

Next, assuming for simplicity that the gas is completely ionized and composed entirely of hydrogen, the number density of protons is given by

$$n \simeq \frac{\rho}{m_p}$$

since  $m_p \simeq m_H$ . From the ideal gas law (Eq. 10.11), the pressure of the gas may be written as

$$P = 2nkT,$$

where  $\mu = 1/2$  for ionized hydrogen and  $m_H \simeq m_p$ . Substituting expressions for the pressure and density into Eq. (11.4), the hydrostatic equilibrium equation becomes

$$\frac{d}{dr}(2nkT) = -\frac{GM_\odot nm_p}{r^2}. \quad (11.5)$$

Making the assumption that the gas is isothermal, Eq. (11.5) can be integrated directly to give an expression for the number density (and therefore the pressure) as a function of radius. It is left as an exercise to show that

$$n(r) = n_0 e^{-\lambda(1-r_0/r)}, \quad (11.6)$$

where

$$\lambda \equiv \frac{GM_\odot m_p}{2kTr_0}$$

and  $n = n_0$  at some radius  $r = r_0$ . Note that  $\lambda$  is approximately the ratio of a proton's gravitational potential energy and its thermal kinetic energy at a distance  $r_0$  from the center of the Sun. We now see that the pressure structure is just

$$P(r) = P_0 e^{-\lambda(1-r_0/r)},$$

where  $P_0 = 2n_0 kT$ .

An immediate consequence of Eq. (11.2) is that in our isothermal approximation the pressure *does not* approach zero as  $r$  goes to infinity. To estimate the limiting values of  $n(r)$  and  $P(r)$ , let  $T = 1.5 \times 10^6$  K and  $n_0 = 3 \times 10^{13} \text{ m}^{-3}$  at about  $r_0 = 1.4 R_\odot$ , values typical of the inner corona. Then  $\lambda \simeq 5.5$ ,  $n(\infty) \simeq 1.2 \times 10^{11} \text{ m}^{-3}$ , and  $P(\infty) \simeq 5 \times 10^{-6} \text{ N m}^{-2}$ . However, as we will see in Section 12.1, with the exception of localized clouds of material, the actual densities and pressures of interstellar dust and gas are much lower than those just derived.

Given the inconsistency that exists between the isothermal, hydrostatic solution to the structure of the corona and the conditions in interstellar space, at least one of the assumptions made in the derivation must be incorrect. Although the assumption that the corona is approximately isothermal is not completely valid, it is roughly consistent with observations. Recall that near Earth ( $r \sim 215 R_\odot$ ), the solar wind is characterized by temperatures on the order of  $10^5$  K, indicating that the temperature of the gas is not decreasing rapidly with distance. It can be shown that solutions that allow for a realistically varying temperature structure still do not eliminate the problem of a predicted gas pressure significantly in excess of the interstellar value. Apparently, it is the assumption that the corona is in hydrostatic equilibrium that is wrong. Since  $P(\infty)$  greatly exceeds the pressures in interstellar space, material must be expanding outward from the Sun, implying the existence of the solar wind.

### The Hydrodynamic Nature of the Upper Solar Atmosphere

If we are to develop an understanding of the structure of the solar atmosphere, the simple approximation of hydrostatic equilibrium must be replaced by a set of **hydrodynamic equations** that describe the flow. In particular, when we write

$$\frac{d^2 r}{dt^2} = \frac{dv}{dt} = \frac{dv}{dr} \frac{dr}{dt} = v \frac{dv}{dr},$$

Eq. (10.5) becomes

$$\rho v \frac{dv}{dr} = -\frac{dP}{dr} - G \frac{M_r \rho}{r^2}, \quad (11.7)$$

where  $v$  is the velocity of the flow. With the introduction of a new variable (velocity), another expression that describes the conservation of mass flow across boundaries must also be included, specifically

$$4\pi r^2 \rho v = \text{constant},$$

which is just the relationship that was used in Example 11.2.1 to estimate the Sun's mass loss rate. This expression immediately implies that

$$\frac{d(\rho v r^2)}{dr} = 0.$$

At the top of the convection zone, the motion of the hot, rising gas and the return flow of the cool gas sets up longitudinal waves (pressure waves) that propagate outward through the

photosphere and into the chromosphere. The outward flux of wave energy,  $F_E$ , is governed by the expression

$$F_E = \frac{1}{2} \rho v_w^2 v_s, \quad (11.8)$$

where  $v_s$  is the local sound speed and  $v_w$  is the velocity amplitude of the oscillatory wave motion for individual particles being driven about their equilibrium positions by the “piston” of the convection zone.

From Eq. (10.84), the sound speed is given by

$$v_s = \sqrt{\gamma P / \rho}.$$

Since, according to the ideal gas law,  $P = \rho k T / \mu m_H$ , the sound speed may also be written as

$$v_s = \sqrt{\frac{\gamma k T}{\mu m_H}} \propto \sqrt{T}$$

for fixed  $\gamma$  and  $\mu$ .

When the wave is first generated at the top of the convection zone,  $v_w < v_s$ . However, the density of the gas that these waves travel through decreases significantly with altitude, dropping by four orders of magnitude in approximately 1000 km. If we assume that very little mechanical energy is lost in moving through the photosphere (i.e.,  $4\pi r^2 F_E$  is approximately constant) and that  $v_s$  remains essentially unchanged since the temperature varies by only about a factor of two across the photosphere and chromosphere, the rapid decrease in density means that  $v_w$  must increase significantly (approximately two orders of magnitude). As a result, the wave motion quickly becomes *supersonic* ( $v_w > v_s$ ) as particles in the wave try to travel through the medium faster than the local speed of sound. The result is that the wave develops into a **shock wave**, much like the shock waves that produce sonic booms behind supersonic aircraft.

A shock wave is characterized by a very steep density change over a short distance, called the **shock front**.<sup>10</sup> As a shock moves through a gas, it produces a great deal of heating via collisions, leaving the gas behind the shock highly ionized. This heating comes at the expense of the mechanical energy of the shock, and the shock quickly dissipates. Thus the gas in the chromosphere and above is effectively heated by the mass motions created in the convection zone.

### Magnetohydrodynamics and Alfvén Waves

It should be noted that our discussion of the hydrodynamic equations has failed to account for the influence of the Sun’s magnetic field. It is believed that the temperature structure throughout the outer solar atmosphere, including the very steep *positive* temperature gradient in the transition region, is due at least in part to the presence of the magnetic field, coupled with mass motions produced by the convection zone. **Magnetohydrodynamics**

<sup>10</sup>Recall the discussion of the heliopause at the end of Example 11.2.1.



(usually mercifully shortened to **MHD**) is the study of the interactions between magnetic fields and plasmas. Owing to the great complexity of the problem, a complete solution to the set of MHD equations applied to the outer atmosphere of the Sun does not yet exist. However, some aspects of the solution can be described.

The presence of the magnetic field allows for the generation of a second kind of wave motion. These waves may be thought of as transverse waves that propagate along the magnetic field lines as a consequence of the restoring force of tension associated with the magnetic field lines. To understand the origin of this restoring force, recall that establishing a magnetic field (which is always generated by moving electric charges, or currents) requires that energy be expended. The energy used to establish the field can be thought of as being stored within the magnetic field itself; thus the space containing the magnetic field also contains a magnetic energy density. The value of the magnetic energy density is given by

$$u_m = \frac{B^2}{2\mu_0}. \quad (11.9)$$

If a volume  $V$  of plasma containing a number of magnetic field lines is compressed in a direction perpendicular to the lines, the density of field lines necessarily increases.<sup>11</sup> But the density of field lines is just a description of the strength of the magnetic field itself, so the energy density of the magnetic field also increases during compression. An amount of mechanical work must therefore have been done in compressing the field lines in the gas. Since work is given by  $W = \int P dV$ , the compression of the plasma must imply the existence of a magnetic pressure. It can be shown that the magnetic pressure is numerically equal to the magnetic energy density, or

$$P_m = \frac{B^2}{2\mu_0}. \quad (11.10)$$

When a magnetic field line gets displaced by some amount perpendicular to the direction of the line, a *magnetic pressure gradient* becomes established; the pressure in the direction of the displacement increases as indicated by an increase in the number density of field lines, while at the same time the pressure in the opposite direction decreases. This pressure change then tends to push the line back again, restoring the original density of field lines. This process may be thought of as analogous to the oscillations that occur in a string when a portion of the string is displaced; it is the tension in the string that pulls it back when it is plucked. The “tension” that restores the position of the magnetic field line is just the magnetic pressure gradient.

As with the traveling motion of a wave on a string, a disturbance in the magnetic field line can also propagate down the line. This transverse MHD wave is called an **Alfvén wave**.<sup>12</sup>

<sup>11</sup> Recall that if the electric field is negligible, charged particles must spiral around field lines. This implies that if the charged particles are pushed, they drag the field lines with them; the field lines are said to be “frozen in” the plasma.

<sup>12</sup> Alfvén waves are named for Hannes Olof Gösta Alfvén, (1908–1995), who was awarded the Nobel Prize in 1970 for his fundamental studies in magnetohydrodynamics.



The speed of propagation of the Alfvén wave may be estimated by making a comparison with the sound speed in a gas. Since the adiabatic sound speed is given by

$$v_s = \sqrt{\frac{\gamma P_g}{\rho}},$$

where  $\gamma$  is of order unity, by analogy the *Alfvén speed* should be approximately

$$v_m \sim \sqrt{\frac{P_m}{\rho}} = \frac{B}{\sqrt{2\mu_0\rho}}.$$

A more careful treatment gives the result

$$v_m = \frac{B}{\sqrt{\mu_0\rho}}. \quad (11.11)$$

---

**Example 11.2.2.** Using Eqs. (10.84) and (11.11), the sound speed and Alfvén speed may be compared for the photosphere. The gas pressure at the top of the photosphere is roughly  $140 \text{ N m}^{-2}$ , with a density of  $4.9 \times 10^{-6} \text{ kg m}^{-3}$ . Assuming an ideal monatomic gas for which  $\gamma = 5/3$ ,

$$v_s \simeq 6900 \text{ m s}^{-1}.$$

Notice that this speed is significantly lower than we found using global solar values in Example 10.4.2; apparently, the sound speed is much larger in the Sun's interior.

Taking a typical surface magnetic field strength to be  $2 \times 10^{-4} \text{ T}$ , the magnetic pressure is (from Eq. 11.10)

$$P_m \simeq 0.02 \text{ N m}^{-2},$$

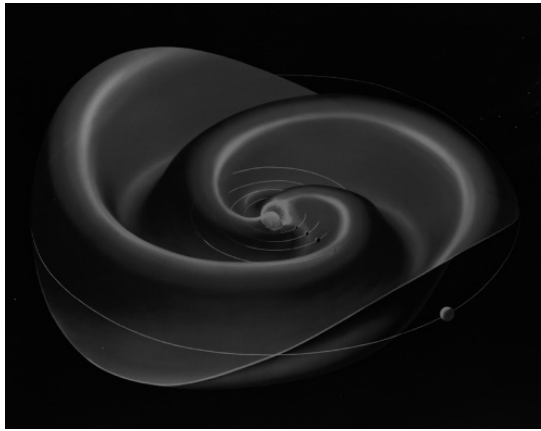
and the Alfvén speed is

$$v_m \simeq 81 \text{ m s}^{-1}.$$

The magnetic pressure may generally be neglected in photospheric hydrostatic considerations since it is smaller than the gas pressure by roughly four orders of magnitude. However, we will see in the next section that much larger magnetic field strengths can exist in localized regions on the Sun's surface.

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Since Alfvén waves can propagate along magnetic field lines, they may also transport energy outward. According to Maxwell's equations, a time-varying magnetic field produces an electric field, which in turn creates electrical currents in the highly conductive plasma. This implies that some resistive Joule heating will occur in the ionized gas, causing the temperature to rise. Thus MHD waves can also contribute to the temperature structure of the upper solar atmosphere.



**FIGURE 11.27** The Sun's rotation creates a spiral pattern in the solar magnetic field in interplanetary space, known as the Parker spiral. The drag produced by the spiraling magnetic field causes angular momentum to be transferred away from the Sun. This diagram shows the heliospheric current sheet that separates regions of space where the magnetic field points toward or away from the Sun. The orbits of the planets out to Jupiter are depicted. (Courtesy of Prof. John M. Wilcox and NASA artist Werner Heil.)

Because of the Sun's rotation, its open magnetic field lines are dragged along through interplanetary space (Fig. 11.27). Since the solar wind is forced to move with the field lines, a torque is produced that actually slows the Sun's rotation. Said another way, the solar wind is transferring angular momentum away from the Sun. As a result, the Sun's rotation rate will decrease significantly over its lifetime. Interestingly, the differential rotation present in the photosphere is not manifested in the corona. Apparently, the magnetic field, which so strongly influences the structure of the corona, does not exhibit differential rotation at this height.

### The Outer Atmospheres of Other Stars

Although this chapter is devoted to our Sun, the most thoroughly studied of all stars, the outer atmospheres of other stars can be investigated as well. For instance, observations indicate that the rotation rates of solar-type stars seem to decrease with age. Furthermore, late main-sequence stars, with their convective envelopes, generally have much slower rotation rates than stars on the upper end of the main sequence. Perhaps winds are transferring angular momentum away from these lower-mass stars as well.

A host of satellites such as EUVE, FUSE, ROSAT, ASCA, XMM-Newton, and the Chandra X-Ray Observatory have also provided us with valuable UV and X-ray observations of other stars. It appears that stars along the main sequence that are cooler than spectral class F have emission lines in the ultraviolet that are similar to those observed coming from the Sun's chromosphere and transition region. In addition, X-ray observations indicate corona-like emissions. These stars are also those for which stellar structure calculations indicate that surface convection zones should exist. Apparently, the same mechanisms that are heating the outer atmosphere of our Sun are also in operation in other stars.